



OYA HYBRID PROJECT



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Prepared by	Grant Little - Anthesis South Africa & Carbon Development Team

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Oya Hybrid Project combines 155 MW of solar PV power, 86.4 MW of wind power, and a 94 MW battery energy storage system (BESS). All power will either be used to feed the grid, charge the BESS, provide power to facility or be throttled to prevent loss. All infrastructure is located on a single site, and combined the system will provide 128 MW of dispatchable power. This greenfield development spans the Western and Northern Cape Provinces, between the towns of Ceres and Sutherland in South Africa.

At the heart of this groundbreaking project are co-located wind turbines, solar PV arrays, lithium-ion phosphate Batteries Energy Storage System (BESS), and an innovative hybrid energy controller. This integrated system efficiently orchestrates all three technologies, providing dispatchable power to the grid through the Hybrid Power Controller (HPC). Renewable energy generated by the solar PV and wind facilities is routed through the HPC, which ensures that all renewable energy is directed to the BESS for storage or dispatched as per the optimum power production and capacity. The HPC optimizes and manages any direct export of solar or wind generation to the grid, and electricity discharged from the facility and delivered to the grid.

A 132kV distribution line of approximately 33km in length feeds the electricity to Eskom's existing Kappa Main Transmission Substation in the Western Cape where it will be distributed into the South African grid. Not only is the Oya Energy Project currently the largest hybrid BESS, renewable energy project of its kind in Africa, it is also globally unique in terms of its technology mix, size, and cost-effectiveness. Its strategic location boasts abundant natural resources and proximity to the grid connection point, is complemented by advanced battery storage capabilities. The infrastructure is summarized as:

- Capacity: 155 MW of solar PV power, 86.4 MW of wind power, and a 94 MW battery energy storage, combined to provide 128 MW dispatchable power
- Technology Mix: Wind, Solar PV, Lithium-Ion Phosphate Batteries
- Transmission Network: Connected to the South African National grid via 132 kV substation
- Construction Timeline: To be fully completed in Q4 of 2025

The situation existing prior to implementation of project activity is that the electricity delivered to the grid would have been generated by the operation of fossil fuel-based power plants. The project activity generates renewable electricity and helps to displace CO₂ emission from fossil fuel-based grid connected power plants.

The project is expected to result in an average annual reduction of approximately 522,354 tCO₂e and a total of 3,656,480 tCO₂e emission reductions over the first crediting period of seven years.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	01-October-2025 to 30-September-2032	VCS	Carbon Check (India) Private Ltd.	Seven (7)
	...			

1.3 Sectoral Scope and Project Type

Sectoral scope¹	01 - Energy industries (Renewable/non-renewable sources)
Project activity type	Renewable Energy Project

1.4 Project Eligibility

1.4.1 General eligibility

As per paragraph 2.1.1 of the VCS Standard, version 4.7, the project is eligible under the scope of the VCS Program because this project activity meets three of the listed criteria, as described in Table 1 below. The project is also eligible under VCS v4.7, Table 2.1, as shown in Table 2 below:

Table 1: Scope of the VCS program

Elements	Justification of project applicability
1) The seven Kyoto Protocol greenhouse gases.	The project aims at reducing greenhouse gas (GHG) emissions by replacing fossil fuel energy supplied to the South African national grid, with clean renewable electricity.
2) Ozone-depleting substances (ODS).	Not applicable

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Project activity applies VCS approved, CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ² which has been approved for use under the VCS Program,
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ³ .
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	Not applicable

Table 2: Excluded Project Activities under the VCS v 4.7 Table 1

Activity	Applicability
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid.	Not applicable to this project activity as it is not a hydroelectric power plant.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	This is a grid connected electricity generation activity using a combination of 155 MW of solar PV power, 86.4 MW of wind power, and a 94

² <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

³ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

	MW battery energy storage (BESS) with a hybrid power controller. Hence the project is eligible and not excluded.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).	Not applicable to this project which is not a waste heat recovery project.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass. The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO ₂ capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves).	Not applicable to this project which does combust biomass.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste containing previously used petroleum products (e.g., used plastics, oils, lubricants).	Not applicable to this project which does not combust fossil fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO ₂ e per year.	Not applicable to this project which does not replace electric lighting.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.	Not applicable to this project which does not install or

Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO ₂ e per year.	replace electricity transmission lines or transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	Not applicable to this project which does not reduce HFC-23 emissions.

As per para 3.8.1 of VCS standard V 4.7, non-AFOLU projects shall complete validation within two years of the project start date & this condition will be fulfilled by this project activity as the start date of the project activity is expected to be 01-October-2025 and validation commenced in 2024. OYA Hybrid Project applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0), which has been approved by Verra for use under the VCS Program for project activities seeking registration with them. ACM0002 is a large-scale methodology and has no capacity limit.

1.4.2 AFOLU project eligibility

This section is not applicable, as the project is not an AFOLU project

1.4.3 Transfer project eligibility

This section is not applicable as the project is not being transferred from another GHG Program nor a CPA seeking registration under the VCS Program.

1.5 Project Design

Indicate if the project has been designed as:

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

This section is not applicable as the project is a single location project.

1.6 Project Proponent

Organization name	Anthesis B.V. (Anthesis')
Contact person	Grant Little
Title	Principal Carbon Developer
Address	Workshop 17, 32 Kloof Street, Cape Town, South Africa
Telephone	+27 21 300 6770
Email	grant.little@anthesisgroup.com

Organization name	Oya Energy (Pty) Ltd
Contact person	Sherwin Harris
Title	Chief Executive Officer
Address	Building 1 Country Club Estate, 21 Woodlands Drive Woodmead, Johannesburg, Gauteng, 2191
Telephone	+27 10 612 0700
Email	sherwin.harris@oya-sa.com

1.7 Other Entities Involved in the Project

There are no other entities involved in the development of this project.

1.8 Ownership

Oya Energy (Pty) Ltd is the project owner of OYA Hybrid Project. They have the legal right to control and operate the project activity. The ownership can be verified through following documents:

- Lease Agreements
- Servitudes Agreements
- Landowner's contracts
- Implementation Agreements
- Power Purchase Agreement

1.9 Project Start Date

Project start date	01-October-2025
Justification	According to Section 3.8 in the VCS Standard Version 4.7, the project start date of a non-AFOLU project is the date on which the project begins generating GHG emission reductions or carbon dioxide removals. In this case the start date of project activity is the Commercial Operation Date (COD), on which the plant first starts to feed electricity into the national grid. As such, that is the day on which the project starts generating emission reduction credits. This has been taken as per the Power Purchase Agreement as the date above when COD is planned.

1.10 Project Crediting Period

Crediting period	☒ <i>Seven years, twice renewable</i> ☐ <i>Ten years, fixed</i> ☐ <i>Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)</i>
Start and end date of first or fixed crediting period	01-October-2025 to 30-September-2032

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals (ERRs) of the project are:

- ☐ < 300,000 tCO₂e/year (project)
☒ ≥ 300,000 tCO₂e/year (large project)

The table below shows the estimated annual GHG emissions for the first renewable crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-October-2025 to 31-December-2025	130,589

01-January-2026 to 31-December-2026	522,354
01-January-2027 to 31-December-2027	522,354
01-January-2028 to 31-December-2028	522,354
01-January-2029 to 31-December-2029	522,354
01-January-2030 to 31-December-2030	522,354
01-January-2031 to 31-December-2031	522,354
01-January-2032 to 30-September-2032	391,766
Total estimated ERRs during the first or fixed crediting period	3,656,480
Total number of years	7
Average annual ERRs	522,354

1.12 Description of the Project Activity

The Project will have a Contracted Capacity of 128 MW from 2 renewable energy sources (wind and solar PV) feeding into a Battery Energy Storage System (BESS), managed by a hybrid power controller to feed the grid. All technologies will be monitored and controlled through an automatic generation Hybrid Power Plant Controller. The Project will connect to the national grid via an onsite 33/132kV substation through a series of underground and overhead cables connecting the assets to the delivery point, stepping the voltage to 132kV. A 132kV distribution line of approximately 33km in length feeds the electricity to Eskom's existing Kappa Main Transmission Substation in the Western Cape where it will be distributed into the South African grid.

The Project has two main access points, one from the North which will be used to access the wind facility and one in the South to access the solar and BESS facilities. The Project will be equipped with state-of-the-art meteorological instruments, monitoring onsite wind and solar resources to allow for forecast optimisation for power generation. Two on-site met masts will be placed between wind turbine generators. Measurement equipment within the PV facility will monitor solar resource data. A schematic diagram of the facility is shown in Figure 1 below.

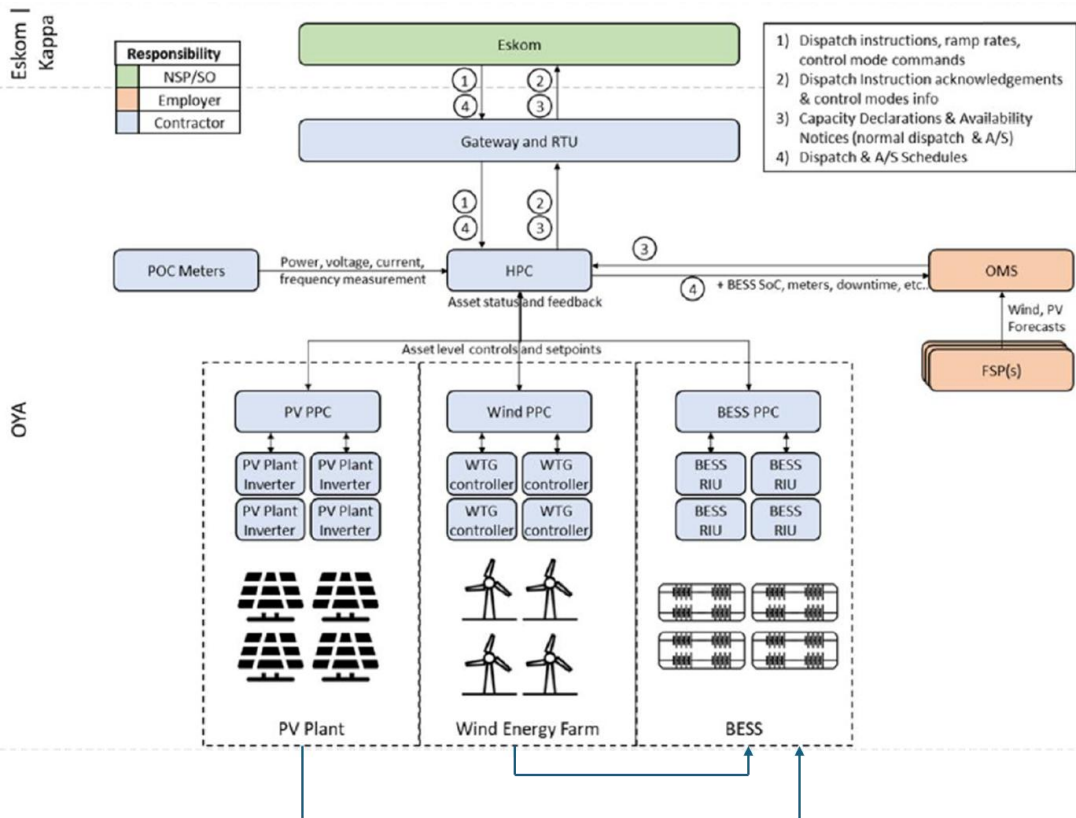


Figure 1: High level schematic diagram of Oya Power Facility

Table 3: Technical specifications for project technologies

TECHNOLOGY	SUPPLIER	CAPACITY	LOAD FACTOR	LIFESPAN
PV Modules	JA Solar (imported) Local (Wip)	155MWp	25% (Approximately, depending on weather conditions)	25 years
PV Inverters	Sungrow	1+X series (4400kw) ~ 30 modular inverters	97%	25 years

Wind Turbines	Goldwind	86.4 MW	35% (Approximately, depending on weather conditions)	25 years
BESS	Sungrow	94 MW Installed	85%	25 years – with a refurbishment after 12 years
Hybrid Power Plant Controller	ETAP		N/A	25 years
Transformers	Actom	2 X 160MVA	N/A	25 years
OMS (Oya Management System)	G7	N/A	N/A	25 years

The proposed hybrid BESS facility will include PV fields (arrays) comprising multiple PV modules (up to 155MW), Wind farm (86.4 MW) and a Battery Energy Storage Facility (BESS) (94 MW storage capacity), regulated by the Hybrid power plant controller. In summary, the proposed solar PV component of the energy facility will be the bulk of the generating facility and the associated infrastructure will include the following components:

Solar PV arrays:

- The facility will include PV fields (arrays) comprising multiple PV modules. The PV modules are arranged in rows and columns, some of which may require levelling of the terrain and associated slope stabilisation measures.
- Each PV module is approximately 2.5m long and 1.2m wide and mounted on supporting structures above ground. The final design layout to be informed by respective specialist walk-through assessments, to confirm solar structures will be primarily north-south facing tracking the sun east-west, using mono-facial, single axis tracking.
- The Solar PV arrays have a planned lifespan of 25 years, even though the PPA duration is 20 years.

Wind Energy Facility:

- The facility will include 18 x 4.8MW Goldwind wind turbines. The wind turbines will be arranged across the site with a total development footprint of less than 11.5ha and will include associated infrastructure such as access roads and electrical distribution lines.
- The turbines (nacelle) will be mounted on a tower about 100m high. Each turbine will have 3 blades with a diameter of 136m.
- The WEF has a planned lifespan of 25 years, even though the PPA is for 20 years.

On-site 33/132kV substation and BESS is the heart of the project and the primary technology applied:

- The construction of one (1) new on-site substation with a voltage capacity of 33/132kV, occupying an area of approximately 4ha. The 33kV yard portion of the substation forms part of the project infrastructure will occupy an area of up to approximately 2ha.
- A new 132kV overhead power line will feed the electricity generated by the proposed facility into the national grid.
- A BESS facility will be located next to the on-site 33/132kV substation, or in between the PV arrays. The BESS would cover an area of up to approximately 7.8ha. The storage capacity of the BESS will be up to 94MW, and the type of technology will be either Lithium Ion or Sodium-Sulphur.
- The BESS has a planned lifespan of 25 years, with a refurbishment planned after 12 years,

Hybrid Power Plant controller:

- Depending on the availability of renewable energy at each time increment, the hybrid power plant controller will optimally meet the contracted energy demand requirements of the PPA either directly from the solar PV or wind, from the BESS, or from a combination of these.
- The HPC is the core of the integrated system and ensures that the contracted PPA dispatchable power capacity is met. It only has the possibility of operating in one of 3 states. The switching and optimization of the system operating within and between these 3 states is done in real time and is SCADA controlled,
-
- Operating State 1: When the combined output of wind and PV significantly exceeds the dispatch power, the BESS charges to absorb the surplus energy. Once the BESS is fully charged, any additional excess energy will result in curtailment.
- Operating State 2: When the combined output of wind and PV fluctuates around the dispatch power, the BESS must frequently charge and discharge to smooth out these fluctuations.
- Operating State 3: When the combined output of wind and PV is significantly lower than the dispatch power, the BESS discharges to compensate for the energy shortfall. If the BESS becomes depleted, the system will be unable to meet the dispatch power requirement, potentially leading to penalties from the grid operator.
- Metering of the system, as per the PPA and also the approved methodology, is at the grid tie in point and the single Eskom meter and single PP check meter at the substation are used to determine the resultant energy and emission reductions. There are no separate feeds to the national utility and the hybrid power project only has a single tie in point to the national utility (Eskom) as the plant is designed as a unit.

Operation and Maintenance (O&M) buildings, for use during the operational phase will be a single-storey building, included in the footprint of the substation (namely within the 4ha footprint). This will form the central control room and base of operations for the project.

Secondary infrastructure:

- Water from available local sources, new or existing boreholes, to be conveyed to the construction batching plant via temporary above-ground pipelines (approximately 50cm in diameter) The necessary approvals from the Department of Water and Sanitation (DWS) for sourcing and temporary on-site storage was obtained.
- Temporary on-site storage of approximately 80m³ diesel tanks during the construction phase.

- New internal roads and infrastructure to facilitate access across the site (between PV arrays, BESS and WEF).

Access roads:

- The existing gravel road, linked to the R356, is the only access road to the proposed facility as it is an existing road and allows direct access to the site.
- During construction, the facility would be accessed via the above-mentioned existing gravel road which has been upgraded and is currently used as an access route to renewable energy facilities on neighbouring properties. Minor road strengthening will take place within the existing road reserves.
- New roads of 4 to 8m wide (with associated servitudes / reserves for above- or below-ground cable installation and maintenance) would be constructed between some of the PV arrays to facilitate access throughout the facility for the construction and operational phases. These are included in the overall development footprint.

Fencing and lighting for safety:

- Fencing will surround the entire facility (for security purposes) and will be approximately 2m high.
- External lighting to facilitate security and O&M on site.

1.13 Project Location

The Project straddles both sides of the Western and Northern Cape provincial boundary in a region colloquially known as the Ceres-Karoo. About two thirds of its infrastructure lies in the Western Cape, within the Witzenberg Local Municipality and Cape Winelands District Municipality. The remainder (eastern/northern part of the Project) is in the Northern Cape, Karoo Hoogland Local Municipality and Namakwa District Municipality. The site coordinates are as follows:

OYA ENERGY: APPLICATION SITE

COORDINATES AT CORNER POINTS (DD MM SS.sss)

POINT	SOUTH	EAST
1	S32° 52' 11.464"	E20° 12' 21.280"
2	S32° 54' 28.981"	E20° 13' 34.753"
3	S32° 54' 34.507"	E20° 12' 20.205"
4	S32° 55' 13.194"	E20° 12' 15.385"
5	S32° 56' 3.895"	E20° 12' 26.330"
6	S32° 56' 12.929"	E20° 11' 16.988"
7	S32° 55' 38.363"	E20° 9' 20.611"
8	S32° 54' 56.533"	E20° 9' 15.790"
9	S32° 54' 45.536"	E20° 8' 55.471"
10	S32° 54' 18.000"	E20° 9' 11.192"
11	S32° 54' 13.557"	E20° 9' 23.325"
12	S32° 53' 59.701"	E20° 9' 41.082"
13	S32° 53' 54.356"	E20° 9' 40.992"
COORDINATES AT CENTRE POINT (DD MM SS.sss)		
POINT	SOUTH	EAST
19	S32° 54' 25.035"	E20° 11' 16.191"

Figure 2: Geodetic co-ordinates for site

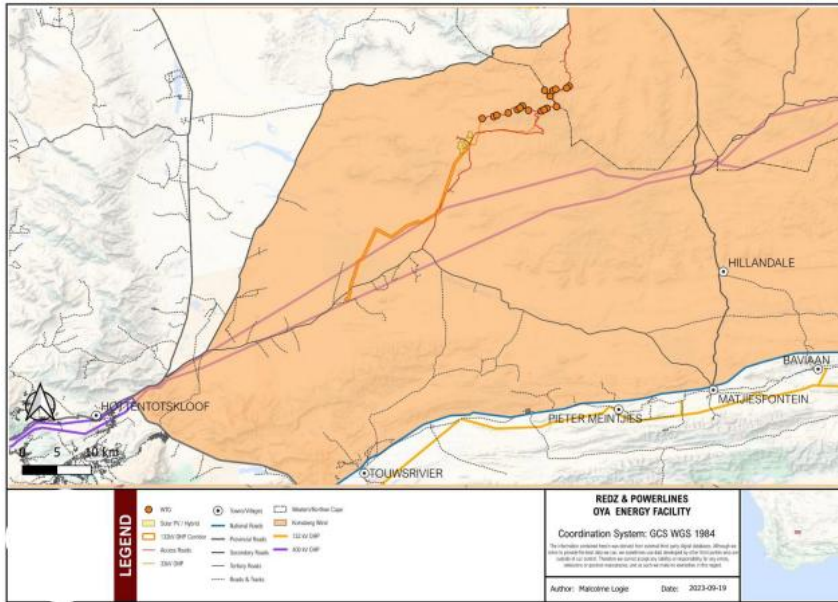


Figure 3: Schematic of location of project

1.14 Conditions Prior to Project Initiation

The electricity generation conditions in South Africa, prior to including this Hybrid BESS plant into the national grid, are described under Section 3.4 (Baseline Scenario).

Although this is not an AFOLU project, the environmental conditions are discussed as this is a Greenfield project and is established on areas previously zoned as agricultural land. The primary purpose of the project is to supply renewable and reliable electricity to South African consumers, and as such, it is not implemented with the sole aim of generating GHG emissions for the purpose of their subsequent reduction, removal, or destruction.

Ecosystem type and Vegetation

The area is characterized by natural vegetation, dominated by Karoo and Fynbos shrubland that is interspersed with natural grassland. Two vegetation types were observed in the site, namely the Koedoesberge-Moordenaars Karoo which consists of low succulent scrub, dotted by scattered tall shrubs, and the Central Mountain Shale Renosterveld with tall shrubland and large suites of non-succulent karoo shrubs. The distribution of vegetation types observed in the project site characterize it as a mosaic of the Succulent Karoo and Fynbos biomes. This ecosystem type shows a high probability of being a habitat to reptile species such as the Karoo Dwarf Tortoise which is a near-threatened species, and floral species such as the *Quaqua mammillaris* which is a protected plant species under the Cape Nature and Environmental

Conservation Ordinance. The leopard (*Panthera pardus*), listed as a Vulnerable species occurs in nearby sites.

Current and Historical Land-use

The proposed project site was previously used for low-intensity grazing; however, the property is no longer actively used for agricultural activities.

Climate

The climate is described as arid with annual rainfall of just over 200 mm that is likely in mid to late winter. The area experiences hot summers, with annual average temperatures ranging from a minimum of 8.9 °C in winter, to a maximum of 34.6 °C in the summer,

Topography

The study area is situated in an area with a gently to moderately sloping topography, with an overall landscape that varies from level to steep. The southern edge of the site is on a mini scarp slope that rises from the plains to approximately 940 – 1120m above sea level. The proposed project site will have its development concentrated on the lower lying areas which are more suitable for the solar panels, whilst the wind turbines will be located along the ridges.

Hydrology

According to the Surface Water Impact Assessment Report, the project site consisted of several ephemeral tributaries with riparian vegetation and episodic drainage lines (EDLs) without riparian vegetation. Since they form part of the drainage system feeding into the Ongeluks River, they trigger the requirement for protection during the construction and operational phases of the project.

Soils

The specialist studies conducted for the project site found that the soils are predominantly shallow and rocky on underlying rock. These attributes of the soils affect the agricultural potential of the site, deeming it unfavourable for crop production.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project activity is in compliance with all the relevant statutory and regulatory laws applicable in the host country.

Relevant laws, statutes, and regulatory frameworks summarized as below:

Table 4: Relevant laws, statutes and regulatory frameworks

Constitution of the Republic of South Africa (Act 108 of 1996)	The Bill of Rights, in the Constitution of South Africa (No. 108 of 1996), gives everyone the right to a non-threatening environment and requires that reasonable measures are applied to protect the environment. This protection encompasses preventing pollution and promoting conservation and environmentally sustainable development. These principles are embraced and given further expression in NEMA.
National Development Plan, Agenda 2030	The NDP envisages that, by 2030, South Africa will have an energy sector that provides reliable and efficient energy service at competitive rates, that is socially equitable through expanded access to energy at affordable tariffs and that is environmentally sustainable through reduced pollution.
Environmental Conservation Act (No. 73 of 1989)	The ECA provides for the effective protection and controlled utilisation of the environment and for matters incidental thereto.
National Environmental Management Act, (No. 107 of 1998)	NEMA requires adherence to the principles of Integrated Environmental Management (IEM) in order to ensure sustainable development, which, in turn, aims to ensure that environmental consequences of development proposals be understood and adequately considered during all stages of the project cycle and that negative aspects be resolved or mitigated, and positive aspects enhanced.
National Water Act (No. 36 of 1998)	Provides the legal framework for the effective and sustainable management of our water resources.
Conservation of Agricultural Resources Act (No. 43 of 1983)	CARA provides for control over the utilization of the natural agricultural resources in order to promote the conservation of the soil, water

	sources and vegetation and the combating of weeds and invader plants; and for matters connected therewith.
Occupational Health and Safety Act (No. 85 of 1993)	OHSA requires the employer to bring about and maintain, as far as reasonably practicable, a work environment that is safe and without risk to the health of the workers. This means that the employer must ensure that the workplace is free of hazardous substances, such as benzene, chlorine and micro-organisms, articles, equipment, processes, etc. that may cause injury, damage or disease. Where this is not possible, the employer must inform workers of these dangers, how they may be prevented, and how to work safely, and provide other protective measures for a safe workplace.
The Basic Conditions of Employment Act (No. 75 of 1997)	The BCEA intends to give effect to the right to fair labour practices referred to in section 23(1) of the Constitution by establishing and making provision for the regulation of basic conditions of employment; and thereby to comply with the obligations of the Republic as a member state of the International Labour Organisation; and to provide for matters connected therewith.
Electricity Regulation Act (Act 4 of 2006) and the Electricity Regulation on New Generation Capacity R399 of 4 May 2011, as amended by R1366 of 04 November 2016	The ERA establishes a national regulatory framework for the electricity supply industry. It designates the National Energy Regulator of South Africa (NERSA) as the custodian and enforcer of this framework. The act covers licensing, registration and regulation of electricity generation, transmission, distribution, trading, import and export. The Regulations provide for the procurement of new generation capacity derived by organs of state, from renewable energy sources and co-generation; base load, mid-merit load and

	peak load new generation capacity; and cross border projects.
Integrated Resource Plan (IRP2019), and the draft IRP 2023	The IRP is the electricity infrastructure development plan, of the Department of Energy, based on the least-cost electricity supply and demand balance, taking into account security of supply and the environment (minimize negative emissions and water usage).
Carbon Tax Act 15 of 2019 and supporting regulations	Provide for VCU's to be used to offset carbon tax liability.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The sustainability objective of the project lies in its capacity to reduce greenhouse gas emissions. Over the seven-year crediting period, renewable twice, the project anticipates an average annual emission reduction of 522,354 tons of CO₂ equivalent, totalling 10 447 086 tons of CO₂ equivalent over the project lifetime. This substantial reduction aligns with Sustainable Development Goal (SDG) 13 (Climate Action), specifically addressing the urgent need to combat climate change and its impacts. Monitoring will be undertaken as per the plan in this Project Description, which involves the monitoring of generated renewable energy in MWh/year. This aligns with South African SDG 7.2 which aims to increase the renewable energy within the country. The project displaces approximately 544,119 MWh per annum of emission-intensive grid electricity through the generation of low emission solar and wind electricity. Furthermore, aligned with SDG 8.5, the project will promote sustained, inclusive and sustainable economic growth as well as full and productive employment and decent work for all. This will be achieved through 250 – 1000 jobs being created during the construction phase and approximately 72 permanent jobs during the ongoing operation of the project into the future. Given the high unemployment rate in the surrounding communities, the employment opportunities created by this project will upskill job seekers in the local community, allowing them to gain experience and transferable skills, enhancing their prospects for future employment. This is how the project contributes to the SDG 8.5 goal for sustained decent work and socio-economic growth.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7,2,1A1 (South African SDG Indicators) – Amount of renewable energy at annual operating capacity	Implemented activities will increase the amount of renewable energy in South Africa	Displaces approximately 544, 119 MWh per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 10 900 GWh of emission intensive grid electricity through the generation of low emission renewable electricity.
2)	13.0	Tonnes of greenhouse gas emissions avoided or removed	The total GHG emissions of the country will be reduced by the ERs and REMs that will no longer be contributing to the country's total GHG emissions. It is expected that the number of GHG emissions will be reduced annually as the project will be verified, confirming that these are no longer emitted to the atmosphere.	Displaces approximately 522, 354 tCO ₂ e per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 10, 447, 000 tCO ₂ e of emission intensive grid electricity through the generation of low emission renewable electricity over the 20-year project lifetime.

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
3)	8.5	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	With total unemployment in South Africa having risen from 24.9% to 33.5% over the past decade (2012 to 2022) this SDG target remains a high priority for the country. The country has a target to, “By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.”	During the construction phase there will be between 250 to 1000 job opportunities created, and during operation there will be approximately 72 jobs. Furthermore, the project has committed to contributing more than 40% of the capital expenditure to local content (procurement of South African goods and services).	The project planned to create between 250 and 1000 job opportunities during the construction phase (from domestic labour) and a further 72 permanent jobs during operations. The Operational jobs will be ongoing, but the construction jobs would only be during the construction phase and will phase out over a period once the plant is operational. Over the 20-year operational period, 1% of revenue will be dedicated to local communities through socio-economic initiatives.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

In accordance with the approved methodology ACM0002, no leakage emissions are considered in the project activity.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

No further Information submitted.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

The table below describes the stakeholder identification process followed.

Stakeholder Identification	An extensive stakeholder engagement process was conducted during the Environmental Impact Assessment (EIA) phase of the project development, as per the NEMA EIA Regulations (2014, as amended). Prior to the construction of the Oya Hybrid Project, a Public Participation Process (PPP) was conducted as part of the EIA to engage interested and affected parties (I&APs). The purpose of these consultations was to inform the stakeholders about the proposed project – including its design, planned implementation, potential risks, and anticipated benefits – and to provide an opportunity for feedback and input. Stakeholders were registered via in-person visits, SMS, email or telephonically. Their interest in the project activity was requested and all information regarding the project was shared with them in their preferred medium. The following measures were implemented to identify stakeholders: - Site Notices in English and Afrikaans were placed along the entrance road to the Gatsriver guest farm from 09 October 2020. - Advert placed in the local newspaper (the Noordwester) on 16 October 2020 - Notifications regarding the EIA process and availability of project reports for public review sent via email/SMS notifications on 09 October 2020. - Due to COVID-19 restrictions, public meetings could not be held when the PPP process was initiated in 2020. However, an I&AP register was compiled, and the stakeholders were invited to the in-
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person meetings held in 2024. The following public meetings were held by the Community Liaison Officer (CLO) based in Matjiesfontein:

- Laingsburg – 15 July 2024 – 50 people in attendance
- Touws River – 16 July 2024 – 120 people in attendance
- Ceres – 17 July 2024 – 60 people in attendance
- Sutherland – 18 July 2024 – 80 people in attendance

All interested and affected party (I&AP) details were compiled into a database which includes contact details, dates and details of consultations, and a record of all issues raised, and responses provided. This database will be updated on an on-going basis. Key stakeholders include representatives from:

- Breede Gouritz Catchment Management Agency
- CapeNature
- Cape Winelands District Municipality
- Ceres-Tankwa Karoo Agricultural Association (CTKAA)
- Department of Agriculture, Forestry and Fisheries
- Department of Agriculture
- Department of Agriculture, Land Reform & Rural Development
- Department of Cultural Affairs & Sport
- Department of Environment, Forestry and Fisheries - Biodiversity
- Department of Environment, Forestry and Fisheries – Air Quality
- Department of Environmental Affairs & Development Planning
- Department of Mineral Resources & Energy (DMRE)
- Department of Transport And Public Works
- Department of Water And Sanitation
- Endangered Wildlife Trust (EWT)
- ESKOM
- Heritage Western Cape (HWC)
- National Defence Force
- SA Civil Aviation Authority (SA CAA)
- SANRAL - Western Region
- SENTECH
- South African Astronomical Observatory (SAAO)

	- South African Local Government Association - South African Weather Service - Southern African Large Telescope (SALT) - Square Kilometre Array (SKA) - TELKOM - Transnet Freight Rail - Western Cape Department of Environmental Affairs and Development Planning (WC DEE&DP) - Air Quality - Wildlife and Environment Society South Africa (WESSA) - Witzenberg Local Municipality (Environmental and Air Quality)
Legal or customary tenure/access rights	The landowner and/or occupant of the property / farm portion, on which the project activity occupied, has been notified. The notification has been submitted to the Department of Forestry Fishery and the Environment (DFFE) for consideration together for decision-making. In addition, Landowner Consent has been obtained for the landowner of the property / farm portion and is available.
Stakeholder diversity and changes over time	The stakeholders fall within the Witzenberg Local Municipality; this is located within the Cape Winelands District Municipality and the Western Cape Province. This area has an unemployment rate of approximately 20.9% and the following sectors form the basis of the municipal economy: • Agriculture, forestry and fishing • Finance, insurance • Real estate and business services • Wholesale and retail trade, catering and accommodation The racial diversity of stakeholders comprises of 69% coloured, 17.9% black African and 8.95 % white. The primary language spoken is Afrikaans (81.22%) followed by Sesotho and isiXhosa. English is widely spoken and understood as a second language. Minimal changes are expected in the make-up of this group throughout the lifetime of this project.
Expected changes in well-being	While there are minimal people living close to the site, any minor negative impacts to the well-being of the stakeholders are mainly limited to the construction period. This includes factors like

	annoyance, dust, traffic, roadworks and noise. There will also be positive impacts such as economic inputs to the local economy. After construction is completed, the impacts will be mostly positive, and include job creation, skills development, improved infrastructure and socio-economic stimulation [1].
Location of stakeholders	The stakeholders that are most likely to be impacted by the project reside in the surrounding towns and communities of [1]: • Alfred's Hamlet; • Ceres; • Op-die-Berg; • Prince Wolsley • Tulbagh • Agter-Witzenberg; • The northern portion of the Breede River Valley (Het Land van Waveren) • The Koue Bokkeveld • The Warm Bokkeveld • Sutherland • Matjiesfontein • Laingsburg • Touws River.
Location of resources	The project is located on privately owned agricultural land which is leased from the landowner and therefore no stakeholders have ownership or customary access rights to the project property.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	15-18 July 2024, October – November 2024 (and also previous engagements described in this document, and an ongoing engagement process)
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Stakeholder engagement process	An extensive stakeholder engagement process formed an integral part of the Environmental Impact Assessment (EIA) and was carried out during the project design phase from August 2020 and communication has been upheld since then [1]. This included stakeholder identification, development of an I&AP database and public meetings as set out in section 2.1.1. above. The purpose of the public meetings was to provide information about the project and to provide stakeholders with a platform to provide input to the project planning process and allow for stakeholders to engage and provide an opportunity to influence the design and implementation of the project. Communication was carried out in a culturally appropriate and inclusive manner by locally appointed Community Liaison Officers (CLO) For the carbon component of the project, additional stakeholder consultation was carried out in the second half of 2024, specifically in relation to the VCS requirements, built on this existing stakeholder engagement process. These stakeholder engagements were led by an independent external consulting company and employees to facilitate communication between the communities, the project developers and Anthesis Group. All stakeholders in the registered I&AP database were contacted directly via phone calls and/or emails and/or SMS, during the months of October and November 2024 and notices were printed out and put up at public locations in the vicinity to the project area. Figures 4 and 5 show the email sent out to stakeholders in October 2024 as well as the newspaper article that was published in the local newspaper outlet, Witzenburg Herald, informing the greater public about the carbon project.
Consultation outcome	All issues raised during the public consultation process have been included in the issues and responses report which forms part of the EIA [1]. All issues that were recorded have been resolved. The main issues raised were in relation to the use of the roads leading to the project site, employment opportunities and the digging of trenches for soil nomination. These issues were amicably resolved through consultations with the construction company to fill up the trenches and regular maintenance of the roads leading to the project site. A neighbour did raise a concern about dust from construction causing coating on certain plants locally and the

	aesthetics of the area being affected with roads being built. A senior member of the carbon team had a telephone call with her on 4 November 2024 and heard and listened to her concerns. There were no comments directly related to the carbon revenue, and it was explained that the additional revenue could support additional initiative of this type. The dust suppression had been handled in the construction and EIA process and the PP was aware of the neighbour's concerned and had also met with her previously. No further action was needed, apart from the dust suppression requirements to be followed. To date, no other issues or comments have been received in relation to the carbon project or the VCS process.
Ongoing communication	Ongoing communication will be via direct electronic communication to stakeholders recorded in the registered I&AP database, as well as public notices placed in public locations in the surrounding communities. In addition, the comments section on the Anthesis website will remain active throughout the lifetime of the project. Furthermore, ongoing liaison and communication between the project owner and neighbouring communities is administrated by a locally employed social facilitator (CLO). In addition, regular face-to-face meetings and monitoring visits are conducted by the project owner.
Stakeholder input	All comments received during the EIA and stakeholder consultation have been carefully considered and addressed in the EIA and project implementation. No issues or comments received were directly related to the VCS validation and verification process, therefore no updates to the project design have been considered necessary. No updates beyond the initial project design which was presented at Validation were considered necessary. Stakeholder comments will continue to be monitored to considered for future revision to the project design, operation and implement.



Figure 4: Advertisement in Witzenberg Herald (Oct 2024)

Oya Renewable Energy Project - stakeholder consent

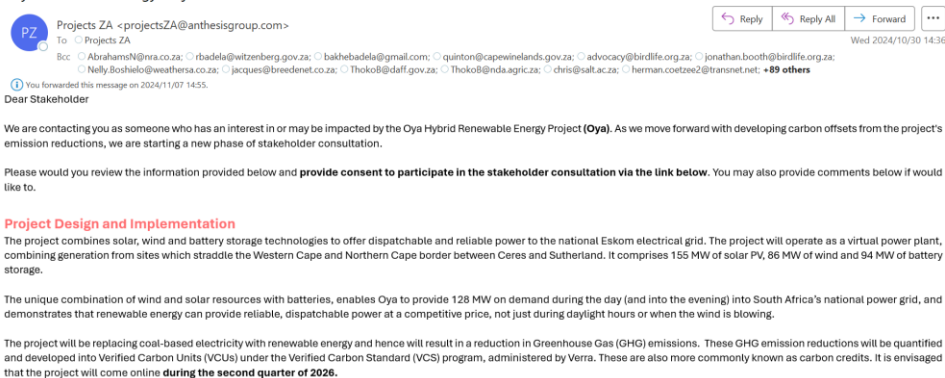


Figure 5: Email sent to stakeholders regarding carbon project (Oct 2024)

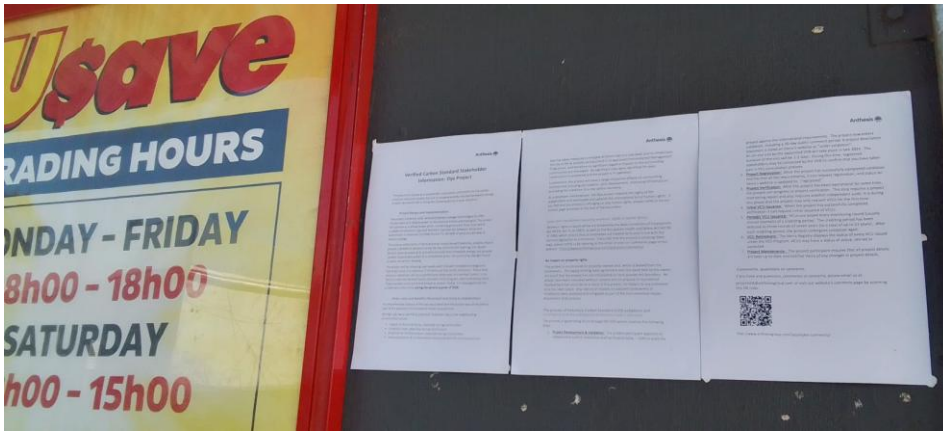


Figure 6: Public notice about carbon project at local shopping centre in Touws River



Figure 7: Public notice on the municipal office noticeboard in Sutherland

2.1.3 Free Prior and Informed Consent

Obtaining consent	The landowner has consented to the project by granting a lease agreement to install the project infrastructure on his property. No IPs (Indigenous Peoples), LCs (Local Communities) or customary rights holders are impacted by the Oya project activities.
Outcome of FPIC	The outcome of the negotiations is that the Oya energy facility will be established on the private land for at least the next 20 years and an agreement is in place with the landowner

2.1.4 Grievance Redress Procedure

Development process	Grievances can be received via two channels: • The grievance portal on the Anthesis website • There are also permanently employed community liaison officers (CLO) stationed in the surrounding communities with the goal of keeping communication up to date with development at Oya. In addition to the CLOs, the project proponent has an in-house team that will liaise with the community for the entire lifetime of the project. Any grievances received will be handled on a case-by-case basis by Anthesis in consultation with the Oya project team.
Grievance redress procedure	Any grievances, complaints or conflicts from stakeholders would be captured via a dedicated feedback form on the Anthesis website, or from the other channels mentioned above. The Grievance Redress process consists of: • Receiving the issue from a stakeholder, either electronically (via email/telephonically) or during public meetings • Assigning it to the relevant manager/team • Responding to the stakeholder (which may include telephonic, written or face to face communication) in an attempt to understand and resolve the issue in good faith Recording the outcome of the interaction. • Grievances that cannot be resolved amicably in this manner will be referred to mediation by a neutral third

	party and failing resolution shall be raised to either arbitration or courts of law in South Africa. Stakeholders have an ongoing opportunity to log into a dedicated feedback form on the Anthesis website in order to leave comments and engage with the project team. This website will be part of the ongoing and continuous engagement between the project and stakeholders. Comments received will be logged and taken into account in project operation and future verifications.
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2.1.5 Public Comments

Comments received	Actions taken
No comments were received during the public commenting period (global stakeholder consultation period on Verra's website).	No changes to the project were required since no public comments were received during this period.
No issues raised or comments received from local stakeholders were directly related to the carbon project or the VCS validation and verification process.	No updates to the project design have been considered necessary,

While no public comments were received from Verra in the initial 30-day Global Stakeholder Comment period, a number of submissions were received from Verra after the initial period had concluded. These comments are addressed below.

Comments received	Actions taken
Comment 1 – Anonymous – submitted 16 May 2025 and published 10 June 2025. It has recently come to our attention that several projects listed below have recently completed their public stakeholder consultation, along the process of applying for registration to generate carbon credits under the VCS. 1. Scatec RMI Hybrid Cluster (https://registry.terra.org/app/projectDetail/VCS/5488)	Stakeholder engagement for the project followed the requirements stipulated in §3.18 of the VCS Standard. This included building on the extensive stakeholder process undertaken during the EIA phase of the project and augmenting this through a renewed review of the stakeholder database to identify and add to it, confirming any stakeholders with legal or customary rights tenure or access rights or any

2. De Aar Hydra Storage
(<https://registry.terra.org/app/projectDetail/VCS/5337>)
3. Oya Hybrid Project
(<https://registry.terra.org/app/projectDetail/VCS/5312>)
4. Umoyilanga Hybrid Project
(<https://registry.terra.org/app/projectDetail/VCS/5272>)
5. Redstone CSP project
(<https://registry.terra.org/app/projectDetail/VCS/5146>)

We have not received any notification from any of the project developers of their intention to pursue registration. As a significantly interested and affected stakeholder, we therefore object to the registration and request further consultation on several issues of concern.

conflicting rights. The stakeholder engagement process included physical notices located in public areas (municipal buildings, local supermarkets etc.) near the project activity, meetings with local stakeholders, advertisements in newspapers, electronic (email) communication, some telephonic calls and a dedicated website link is maintained (<https://www.anthesisgroup.com/za/project-comments/>) that remains open for stakeholder concerns, complaints, grievances and engagement. As part of the operational phase, each project also has a dedicated team of specialists located on site specifically focused on stakeholder engagement. A 30-day Global Stakeholder Consultation (GSC) managed on the Terra registry website was also part of the process as per §3.18.9 – 3.18.13 of the stakeholder engagement process.

The stakeholder engagement process was validated and checked by the independent VVB during the validation process, and included verification of documentation and communication, interviews with the project team members who were involved in the implementation and interviews with a sample of stakeholders themselves.

The comments received from VERRA were anonymised, so it was unsure who the stakeholder was in order to review any possible comments or concerns raised during the engagement process. As part of our ongoing stakeholder engagement (as per §3.18.5 – 3.18.6 of VCS Standard) stakeholder may engage,

	comment, raise a grievance or be added to the stakeholder database. Various mechanisms are available, the primary ones being through the project operational stakeholder teams or via the weblink (https://www.anthesisgroup.com/za/project-comments/). While the stakeholder engagement process sought to identify and include all relevant stakeholders, any stakeholders missed, still have the opportunity to engage and be part of the ongoing process by accessing the weblink above.
Comment 2 – Anonymous – submitted 10 June 2025 FORMAL CONCERN REGARDING REGISTRATION OF IPP PROJECTS UNDER THE VERRA VCS PROGRAMME 1. We refer to the above-subject matter. 2. It has come to our attention that the following Independent Power Producer (IPPs) Projects have been registered or started the process of registering under the Verra Verified Carbon Standard (VCS) programme: 2.1 Scatec RMI Hybrid Cluster (VCS 5488) 2.2 De Aar Hydra Storage (VCS 5337) 2.3 Oya Hybrid Project (VCS 5312) 2.4 Umoyilanga Hybrid Project (VCS 5272) 2.5 Redstone CSP Project (VCS 5146) 3. These projects are being implemented under the Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP) and Renewable Energy Independent Power Producer	The response to the comment has been split into the primary issues raised as follows: • Carbon credits are subject to contractual provisions (§4) • IPP prohibited from registering Environmental Attributes unless prior consent obtained (§5) • Prior authorization required otherwise may be inconsistent with contracts (§6) • Verra to review double counting (including contractual obligations) (§7) • Department request to meet with Verra (§8) Carbon credits are subject to contractual provisions (§4): Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific

Procurement Programme (REIPPPP), managed by the Department of Electricity and Energy through the IPP Office.

4. We wish to formally inform Verra that, in accordance with the Power Purchase Agreements (PPAs), Implementation Agreements (IAs), and associated procurement documents governing these programmes, the ownership and treatment of environmental attributes, including carbon credits and associated claims, are subject to specific contractual provisions.

5. Under these provisions, IPPs are prohibited from registering, monetizing, or otherwise utilizing Environmental Attributes outside of the approved frameworks, unless prior consent has been

obtained from the Department. The Department retains the sole discretion to determine the use, or monetization of Environmental Attributes in terms of the relevant environmental policies.

6. Based on our current understanding, unless prior written authorisation has been obtained from the Department, the IPPs do not hold the exclusive rights to register such environmental attributes under voluntary carbon markets, including the VCS. Accordingly, such registrations, if made without the necessary approval, may be inconsistent with the contractual obligations under which these projects were awarded and developed.

7. Given the potential implications of double-counting, the Department requests that Verra reviews the registration and claims associated with the listed projects, considering the contractual limitations and rights under the REIPPPP and RMIPPPP.

contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. A number of clauses in the Implementation Agreement (IA) and the Power Purchase Agreement (PPA) were referenced in the review and supported this position.

In addition to the above, the monetization of carbon credits resultant from any VCS project, would fall outside of the registration and issuance process, as this would be a commercial agreement between project proponents and shareholders regarding any revenue sharing. The VVB reviewed current contracts between the PPs regarding carbon credits as part of the validation process.

IPP prohibited from registering Environmental Attributes unless prior consent obtained (§5):

Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. Given this lack of contractual specificity, it was deemed that:

- IPPs are not prohibited from registering, monetizing, or otherwise

utilizing Environmental Attributes outside of the approved frameworks,

- the Department does not retain any discretion to determine the use, or monetization of environmental attributes, and
- there are no environmental policies that specifically address these issues except for regulations under the Carbon Tax Act which provide for a process in terms of which project proponents (including project proponents of REIPPPP projects) can seek authorisation to list carbon credits as carbon offsets on the Carbon Offset Administration System. proponent.

Prior authorization required otherwise may be inconsistent with contracts (§6):

Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. Given this lack of contractual specificity:

- IPPs do not need prior authorisation from the Department to register projects under the VCS, indeed a number of REIPPPP projects have been registered and operational for some years, including having transacted VCUs into the carbon. Clearly, therefore no prior authorisation is required.

**Verra to review double counting
(including contractual obligations) (§7):**

In accordance with the VCS Standard (§3.23), project proponents need to demonstrate that there is no double counting, or double selling of any GHG emission reductions. This was confirmed by cross referencing a number of local and international registries, including Renewable Energy Certificates (RECs). Further to this, during the project development process, each project owned provided a signed declaration that no other environmental benefits are sought from the offtake agreements or GHG benefits have been allocated outside of the listed Verra project. These documents formed part of the independent validation by the VVB during the audit process.

In addition, during the project development process, each project owned provided a signed declaration that no other offtake agreements or GHG benefits have been allocated outside of the listed Verra project.

Department request to meet with Verra

(§8): The Project proponent acknowledges that a meeting is requested, and no further action is required currently.

The following confidential documents are referenced in this response:

Independent legal opinions commissioned by the Project Proponents:

- Climate Legal for Anthesis, dated 8 August 2025
- Legal opinion from another IPP, dated 9 September 2025
- Letters to IPPO from Project Proponent (Another IPP) dated 15 July 2025
- Industry Association letter, SAWEA (South African Wind Energy Association) industry letter to the IPP Office, dated 8 August 2025
- Industry Association letter, SACMA (Southern African Carbon Markets Association) industry letter to the IPP Office, dated 4 November 2025
- Letters submitted by other Independent Power Producers to the IPP Office (various dates in 2025), contesting the Department's claim of discretion over environmental attributes and seeking clarity on the contractual basis for such a position
- Implementation Agreement (IA) dated 30 August 2023— Clause 7 (Emission Credits) and Clause 27.2 (No Amendment or Variation)
- Power Purchase Agreement (PPA) dated 30 August 2023

Comment 3 –Elsa Strydom – Submitted 10 June 2025

The response to the comment has been split into the primary issues raised as follows:

1. It has come to our attention that the following Independent Power Producer (IPPs) Projects have been registered or started the process of registering under the Verra Verified Carbon Standard (VCS) programme:

- a) Scatec RMI Hybrid Cluster (VCS 5488)
- b) De Aar Hydra Storage (VCS 5337)
- c) Oya Hybrid Project (VCS 5312)
- d) Umoyilanga Hybrid Project (VCS 5272)
- e) Redstone CSP Project (VCS 5146)

2. These projects are being implemented under the Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP) and Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), managed by the Department of Electricity and Energy through the IPP Office.

3. In accordance with the Power Purchase Agreements (PPAs), Implementation Agreements (IAs), and associated procurement documents governing these programmes, the ownership and treatment of environmental attributes, including carbon credits and associated claims, are subject to specific contractual provisions.

4. Under these provisions, IPPs are prohibited from registering, monetizing, or otherwise utilizing Environmental Attributes outside of the approved frameworks, unless prior consent has been obtained from the Department. The Department retains the sole discretion to determine the use, or monetization of Environmental Attributes in terms of the relevant environmental policies.

5. *Unless prior written authorisation has been obtained from the Department, the IPPs do not hold the exclusive rights to register such environmental attributes under voluntary carbon*

- Carbon credits are subject to contractual provisions (§3)
- IPP prohibited from registering Environmental Attributes unless prior consent obtained (§4)
- Prior authorization required otherwise may be inconsistent with contracts (§5)
- Verra to review double counting (including contractual obligations) (§6)

Carbon credits are subject to contractual provisions (§3): Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers.

In addition to the above, the monetization of carbon credits resultant from any VCS project, would fall outside of the registration and issuance process, as this would be a commercial agreement between project proponents and shareholders regarding any revenue sharing. The VVB reviewed current contracts between the PPs regarding carbon credits as part of the validation process.

IPP prohibited from registering Environmental Attributes unless prior

markets, including the VCS. Accordingly, such registrations, if made without the necessary approval, may be inconsistent with the contractual obligations under which these projects were awarded and developed.

6. Given the potential implications of double-counting, the Department requests that Verra reviews the registration and claims associated with the listed projects, considering the contractual limitations and rights under the REIPPPP and RMIPPPP.

consent obtained (§4): Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. Given this lack of contractual specificity, it was deemed that:

- IPPs are not prohibited from registering, monetizing, or otherwise utilizing Environmental Attributes outside of the approved frameworks,
- the Department does not retain any discretion to determine the use, or monetization of environmental attributes, and
- there are no environmental policies that specifically address these issues except for regulations under the Carbon Tax Act which provide for a process in terms of which project proponents (including project proponents of REIPPPP projects) can seek authorisation to list carbon credits as carbon offsets on the Carbon Offset Administration System. proponent.

Prior authorization required otherwise may be inconsistent with contracts (§5): Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to

environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. Given this lack of contractual specificity:

- IPPs do not need prior authorisation from the Department to register projects under the VCS, indeed a number of REIPPPP projects have been registered and operational for some years within South Africa, including having transacted VCUs into the carbon. Clearly, therefore no prior authorisation is required.

Verra to review double counting (including contractual obligations) (§5): In accordance with the VCS Standard (§3.23), project proponents need to demonstrate that there is no double counting, or double selling of any GHG emission reductions. This was confirmed by cross referencing several local and international registries, including Renewable Energy Certificates (RECs). Further to this, during the project development process, each project owned provided a signed declaration that no other environmental benefits are sought from the offtake agreements or GHG benefits have been allocated outside of the listed Verra project. These documents formed part of the independent validation by the VVB during the audit process.

In addition, during the project development process, each project owned provided a signed declaration that no other offtake agreements or

GHG benefits have been allocated outside of the listed Verra project.

The following confidential documents are referenced in this response:

Independent legal opinions commissioned by the Project Proponents:

- Climate Legal for Anthesis, dated 8 August 2025
- Legal opinion from another IPP, dated 9 September 2025
- Letters to IPPO from Project Proponent (Another IPP) dated 15 July 2025
- Industry Association letter, SAWEA (South African Wind Energy Association) industry letter to the IPP Office, dated 8 August 2025
- Industry Association letter, SACMA (Southern African Carbon Markets Association) industry letter to the IPP Office, dated 4 November 2025
- Letters submitted by other Independent Power Producers to the IPP Office (various dates in 2025), contesting the Department's claim of discretion over environmental attributes and seeking clarity on the contractual basis for such a position
- Implementation Agreement (IA) dated 30 August 2023— Clause 7 (Emission Credits) and Clause 27.2 (No Amendment or Variation)
- Power Purchase Agreement (PPA) dated 30 August 2023

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Anthesis SA team has significant experience in implementing similar project activities, engaging with communities, project management and environmental and social safeguards aligned with the VCS requirements. Members of the project development team have as a collective, in excess of 50 years' experience in the voluntary carbon market. They have been originating, developing and monetizing carbon credits for over 2 decades.

The Oya project proponent has extensive renewable energy project experience in the South African region, including Renewable Power facilities at Klawer, Rietkloof and Roggeveld with their G7 renewable energies partnership. Their Engie team has similar experience across the continent with projects such as Kathu Solar, Xina Solar One, Golden Valley WEF, Aurora and Kookonsies 2. Furthermore, they have project experience in North Africa on projects such as Ras Ghared, Gulf of Suez; Tarfaya in Egypt & Morocco and Kahone Solaire and Kael Solaire in Senegal.

The integrated team is well positioned to implement and operate the facility and deliver the associated GHG emission reductions into the local market.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No significant natural and human induced risks	Environmental and social impacts and risks were documented and addressed in the EIA, SIA and EMPr documents for the WEF, Hybrid BESS and overhead powerlines. Construction activities will cause some changes to the natural environment; no significant adverse risks were identified to stakeholders' wellbeing. Construction activities will cause minor change in landscape

Risks to stakeholder participation	No significant risks identified	characteristics over the localized area effecting the visual quality of the area. Stakeholder engagement process was conducted, and comments received from the stakeholders were properly addressed. No evidence of participation barriers to stakeholders were identified. The process for stakeholder's participation was undertaken as required by the South African EIA regulations in addition to consultation for the VCS project.
Working conditions	No risks identified related to working conditions	All workers and their working conditions are subject to the Basic Conditions of Employment Act of South Africa and labour conditions are covered by periodic inspections and compliance to the South African Department of Labour.
Safety of women and girls	No risks identified related to women and girls identified	The project proponent has taken all measures to ensure safety of women and girls by implementing various policies as per South African labour and anti-discrimination laws.
Safety of minority and marginalized groups, including children	No risks identified	The project proponent ensures safety of minority and marginalized groups, including children in the project site as per the South African labour and anti-discrimination laws.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical	Increased traffic during the construction phase will lead to a degree of increased noise and air pollution from dust.	Dust suppression of gravel roads during the construction phase and regular maintenance of gravel roads during operation phase. Mitigation measures have been put in place to ensure

pesticides and fertilizers)	minimal water pollution to the surrounding watercourses; monitoring plans will also be in place to keep track of any pollution during the construction and operational phases.
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2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁴	Mitigation or preventative measure(s) taken
Discrimination	No risks identified	The project proponents have formal Human Resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Sexual harassment	No risks identified	The project proponents have formal Human Resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Equal pay for equal work	No risks identified	The project proponents have formal Human Resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Gender equity in labor and work	No risks identified	The project proponents have formal Human Resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Forced labor	No risks identified	The project proponents have formal Human Resource and labor policies as per the Basic Conditions of Employment Act of South Africa.

⁴ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Child labor	No risks identified	The project proponents have formal Human Resource and labor policies as per the Basic Conditions of Employment Act of South Africa and the Bill of Rights entrenched within the Constitution of South Africa.
Human trafficking	No risks identified	The project proponents have formal Human Resource and labor policies as per the Basic Conditions of Employment Act of South Africa and the Bill of Rights entrenched within the Constitution of South Africa.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The project proponent adheres to international and national human right laws and the Bill of Rights entrenched within the Constitution of South Africa. Risks related to human rights were assessed during the EIA process.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
Sites of archeological or cultural significance may exist within project boundary	The EMPr incorporated mitigation measures that must be put in place prior to construction to exclude sites of significant cultural or historical (heritage) importance from the project boundary. It was deemed that the project impacts on heritage resources were LOW. It was recommended that the development footprint

be updated in order to accommodate the heritage findings, and that the ECO must make sure the heritage resources are protected should they be identified.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The project is implemented on privately owned agricultural land for which legal land tenure has been secured via lease agreements that are in place. As such, no risks to property rights were identified, and no encroachment on community land is taking place.

2.3.5 Benefit Sharing

As stated in Section 2.3.4, the project does not impact on property rights as the land is privately owned and has been secured via a lease agreement; therefore, there is no Benefit Sharing Agreement.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	Potential risks identified during the EIA authorization process include impacts to surface water features, habitat loss and fragmentation for faunal and flora species.	Ecological specialists delineated development footprint to minimize impacts on favourable habitat for species of conservation concern, and to avoid identified surface water features / watercourses, acknowledging some infrastructure (such as access roads and electricity cables) need to traverse watercourses.
	Increased sedimentation due to soil erosion pose a low-risk significance to the identified ephemeral (non-	Construction works within the

	perennial) tributaries, riparian vegetation and episodic drainage lines [2]. No 'fatal flaws' were identified for freshwater ecosystems. Direct mortality of fauna due to machinery and increased traffic during construction phase. Avifaunal species at risk of collision with solar panels, wind turbines and overhead power lines.	episodic drainage lines and ephemeral tributaries undertaken during dry period to reduce the need for surface water diversion, to limit potential surface water contamination. Exposed soils to be protected during construction, and sediment control devices installed down gradient of construction activities. Storm water management and erosion control plans are in place, with monitoring and maintenance requirements. Access to sensitive areas outside of development footprints not permitted during construction or operations. In order to obtain legal authorization to establish the Oya BESS Hybrid Energy Facility, the requirements of the Environmental Impact Assessment (EIA) Regulations of South Africa were met. Furthermore, to secure funding for the development the International Finance Corporation (IFC) Performance Standards on Environmental and Social Sustainability were followed. A condition of the environmental authorization calls for independent environmental compliance auditing, with reporting to the authorities throughout the construction and operational phases.
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Soil degradation and soil erosion	Potential increased risk of soil erosion due to soil displacement, vegetation clearance and concentrated rainwater run-off from PV panels during operational phase.	Storm water management and erosion control plans are in place, which include on-going monitoring and maintenance. Vehicle traffic limited to established roads and tracks to minimize soil displacement during construction phase. Re-vegetate and re-habilitate disturbed areas, using erosion control mats, silt traps and other mitigation measures post-construction.
Water consumption and stress	Groundwater abstraction for consumption, dust suppression and construction amounted to less than 14% of the maximum sustainable yield of the affected aquifers, therefore registered groundwater uses would not be negatively impacted.[3]	The General Authorization for Water Use, granted by the Department of Water Affairs provides for on-going monitoring, reporting and management throughout the construction and operational phases.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes

☐ No

If yes, list such species and habitats in the table below and provide evidence that the project will not adversely impact these areas.

Species and habitat

The Ecology EIA Study [4] reports that no protected tree species are likely to occur in the study area, while one plant species (*Hoodia gordonii*), protected under the National Environmental Management: Biodiversity Act (Act No 10. Of 2004) (NEM:BA) occurs in the region, but none were found on site during the construction phase. A number of plant species and families occurring on site are protected under provincial legislation (the Cape Nature and Environmental Conservation Ordinance 19 of 1974), but none of these are of conservation concern. However, permits were required by the Provincial authorities to destroy or remove them. A Botanical Search and Rescue Report [5] lists the 1,066 protected plants relocated to suitable areas (outside of the development footprint) prior to construction. Post construction monitoring [6] located 60,13% (641 of the 1066 relocated plants), Of these 641 plants 363 were reported alive and 278 were dead. Thus an average survivorship of 56% was recorded, which is within the estimated 50 - 80% survivorship ratio suggested by Hoare, 2021 [7].

The Ecology EIA Study reports that of the 56 mammal species with distribution ranges across the region, seven (7) are species of concern and protected under national legislation, namely the critically endangered Riverine Rabbit, endangered Black Rhinoceros (arid ecotype), near threatened Honey Badger and Grey Rhebok, and vulnerable Black-footed Cat, Leopard and Cape Fox. However, given the nature of the project development and the fact that these threatened, near threatened or protected mammal species are relatively mobile, the Ecology Specialist reported they are not likely to be significantly negatively impacted by project activities.

A total of 74 reptile species occur in the greater region, of which two species of conservation concern could potentially occur in the project area, namely the near threatened Karoo Dwarf Tortoise, and the protected Armadillo Girdled Lizard [4]. The site also has suitable habitat for a number of frog species, although none are listed or protected species.

	A total of 99 bird species potentially occur in the broader region, of which 39 species are classified as priority species [8]. The Ornithologist raised no objections to the development, provided the proposed mitigation measures were strictly implemented and incorporated into the EMPr. The environmental sensitivity of the WEF development was escalated during construction when an active nest of a Verreaux's Eagle (listed as Vulnerable) was discovered during avifaunal monitoring in September 2023 [9].
Areas needed for habitat connectivity	The development footprint was delineated to exclude sensitive areas to ensure the project footprint does not adversely affect crucial areas for habitat connectivity.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	The development footprint occurs in the Succulent Karoo Biome, in the Rainshadow Valley Karoo Bioregion (Mucina & Rutherford 2006) with two dominant vegetation types; the Koedoesberge-Moordenaars Karoo (across most of the site), and small patches of Central Mountain Shale Renosterveld (occurs on ridge summits). Both vegetation types are listed as "Least Threatened" with less than 1% transformed, and neither is listed in the National List of Ecosystems that are Threatened and need of	Environmental authorization to establish the Oya BESS Hybrid Energy Facility, required extensive Environmental Impact Assessments (EIA) with supporting Specialist studies, The mitigation measures proposed by the Environmental Assessment Practitioners and Specialists are incorporated into the site's EMPr and include the appointment of an Environmental Control Officer (ECO) during construction and operations. Compliance to the EA conditions and EMPr requirements are audited by independent third parties during the construction and operational phases and reported to the authorities.

Areas for habitat connectivity	protection (GN1002 of 2011). However, the Succulent/Nama Karoo vegetation is recognized as a global biodiversity hotspot, sensitive to disturbance, as it can take decades to rehabilitate disturbed areas.	Pre-construction walk-throughs by fauna and flora Specialists were also undertaken to relocate sensitive plant species and to move any individual animals, such as tortoises, prior to construction. Permits to harvest and relocate sensitive flora species were obtained from the respective provincial authorities. Best practice Botanical Search and Rescue and Transplantation Protocol were followed, with monitoring, evaluation and reporting to the authorities.
	The development site is not located within a formally protected area, an important birding area (IBA), or the National Protected Areas Expansion Strategy (NPAES) priority areas. The Cedarberg - Koue Bokkeveld Complex IBA (SA101), the closest IBA is located approximately 34km west of the site, while the Zwartbosch Private Nature Reserve is approximately 40km away [8].	Areas of ecological sensitivity on site must be fenced off to limit movement within the project development footprint.

2.4.2 Introduction of Species

Species introduced	Classification	Justification for use	Adverse effects and mitigation
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No invasive species will be used or introduced at any stage of the project activity.	N/A	N/A	The EMPr require an alien vegetation management plan to be in place for the duration of the operational phase
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Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
No invasive species were identified on the project site.	As per the EMPr, ongoing monitoring for the establishment of alien and invasive vegetation species should be undertaken, specifically at the road crossings and disturbed areas. This forms part of the monitoring as stipulated in the Alien and Invasive Plant Management Plan.

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	A risk of establishment and spread of alien invasive plants due to the clearing and disturbance of indigenous vegetation exists.	Alien and Invasive Plant Management Plan was developed to prevent the introduction of invasive species. In addition, vehicles used in the development site should be regularly washed to avoid the dispersal of seeds of any alien or invasive species on the project site.

2.4.3 Ecosystem Conversion

The project activity is not in any of the categories, ARR, ALM, WRC or ACoGS.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	N/A	Not applicable seeing that this is not in any of the categories, ARR, ALM, WRC or ACoGS



3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	TOOL 01	Tool for the demonstration and assessment of additionality	7.0
Tool	TOOL 07	Tool to calculate the emission factor for an electricity system.	7.0
Tool	TOOL 24	Common practice	3.1
Tool	TOOL 27	Tool for the Investment analysis	14.0

3.2 Applicability of Methodology

The project activity(s) meets each of the applicability conditions of the methodology(s), tools, and modules applied by the project (where applicable), as described below:

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or	The project activity is installation of new grid connected hybrid (solar, wind & BESS) power plants at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield plant) and hence this criterion is applicable as it meets the applicability condition (a).

Methodology ID	Applicability condition	Justification of compliance
	(e) Involve a replacement of (an) existing plant(s)/unit(s); or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid. In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant. The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit,	This is a hybrid power project that consists of 3 major systems: solar PV, Wind and the battery energy storage system (BESS) managed by a hybrid power plant controller. The project is integrating a 94MW BESS. Therefore, it confirms to the criteria of integrate BESS with a Greenfield (renewable) power plant. So, the project meets applicability condition (a).

Methodology ID	Applicability condition	Justification of compliance
	geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the	(a) This is a hybrid power project consists of 3 major systems: solar PV, Wind and the battery energy storage system (BESS). (b) This project is not a capacity addition, retrofit, rehabilitation or replacement, but a greenfield plant. (c) BESS is an integral part of the project design and is a requirement of the project approvals and power purchase agreement. (d) BESS will only be charged from the greenfield renewable energy plants. No grid power will be used to charge the BESS component. No grid power or fossil fuel electricity generator will be used to charge the BESS component.

Methodology ID	Applicability condition	Justification of compliance
	electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	(e) Not applicable to this project as the project does not employ PSP
	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², and all of the following conditions shall apply:	The project activity is an installation of a new grid connected hybrid solar PV & WEF power plants and not Hydro power plant; therefore, this criterion is not applicable for this project

Methodology ID	Applicability condition	Justification of compliance
	(i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project. In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account	The project activity is an installation of a new grid connected hybrid solar PV & WEF power plants and not Hydro power plant; therefore, this criterion is not applicable for this project

Methodology ID	Applicability condition	Justification of compliance
	seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
	In the case of PSP, the project participants shall demonstrate in the VCS Project Description that the project is not using water which would have been used to generate electricity in the baseline.	Not applicable to this project as the project does not employ PSP
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The project activity is installation of new grid connected BESS hybrid system fed from solar PV & WEF power components and does not involve switching from fossil fuel to renewable energy, therefore criterion described in point (a) is not relevant to the project activity. This is a hybrid solar PV & WEF power facility and not a biomass fired plant, therefore criterion described in point (b) is not applicable to the project activity.
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	The project activity is a new grid connected hybrid power plant and not a retrofit, replacement or capacity additions and therefore this criterion is not applicable to the project activity.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Project activity does not entail proposing a new methodology. As such TOOL 01 is mandatory and was therefore applied to prove additionality.
CDM TOOL 01	Once the additionality tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	As per ACM0002 methodology applied by the project activity, the additionality shall be demonstrated and assessed using the TOOL 01.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 07	This tool may be applied to estimate the Operating Margin (OM), Build Margin (BM) and/or Combined Margin (CM) when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	The project activities are a Greenfield solar power generation plants that supplies electricity to the grid. Hence, the “TOOL 07: Tool to calculate the emission factor for an electricity system version 7.0” is applicable.
CDM TOOL 07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under step 2 of the tool are available to the project participants, i.e. option II a and option IIb. If option II a is chosen, the conditions specified in “Appendix 1: Procedures related	Since the project activity is grid connected hybrid solar PV & wind power project this condition is applicable.

to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.

CDM TOOL 07

In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.

The project activity is located in South Africa (a non-Annex I country) and hence the tool is applicable.

CDM TOOL 07

Under this tool, the value applied to the CO₂ emission factor of biofuels is zero.

The project activity does not involve biofuels.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 24	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality” (TOOL 01), the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	The project activity applies TOOL 01 “Tool for the demonstration and assessment of additionality”, therefore, Hence TOOL 24 is applicable for this project.
CDM TOOL 24	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the	The applied methodology ACM0002 does not define a separate approach for conducting of any of common practice test, which are

requirements contained in the methodology shall prevail. different from the ones presented in the tool.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 27	This methodological tool is applicable to CDM project activities and programmes of activities (PoAs) that conduct an investment analysis for the demonstration of additionality and/or the identification of the baseline scenario	The project activity applies the TOOL 01 “Tool for the demonstration and assessment of additionality” and employs an investment analysis, therefore, TOOL 27 is applicable.
CDM TOOL 27	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Applied methodology ACM0002 does not specify any approach for the demonstration of Investment analysis. As per the applied methodology the additionality including investment analysis has been demonstrated as per guidance of the TOOL 01: Tool for the demonstration and assessment of additionality” and TOOL 27: Investment Analysis.

3.3 Project Boundary

The applied methodology ACM0002 version 22, the project boundary includes the power plants and all power plants connected physically to the electricity system that the project power plant is connected to.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

Source		Gas	Included?	Justification/Explanation
Baseline	Grid Connected Electricity Generation	CO ₂	Yes	Main emission source (Grid electricity)
		CH ₄	No	Excluded for simplification. Minor source of emissions. Conservative
		N ₂ O	No	Excluded for simplification. Minor source of emissions. Conservative
		Other	No	No other GHG emissions relevant
Project	Greenfield Solar Photovoltaic electricity generation Project Activity	CO ₂	No	No CO ₂ emissions from project activity. Minimal taken as negligible
		CH ₄	No	Excluded. Minimal taken as negligible
		N ₂ O	No	Excluded. Minimal taken as negligible
		Other	No	No other GHG emissions
	Greenfield Wind electricity generation Project Activity	CO ₂	No	No CO ₂ emissions from project activity. Minimal taken as negligible
		CH ₄	No	Excluded. Minimal taken as negligible
		N ₂ O	No	Excluded. Minimal taken as negligible
		Other	No	No other GHG emissions
	Charging of BESS using electricity from the renewable energy generation plants.	CO ₂	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		CH ₄	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		N ₂ O	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		Other	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.

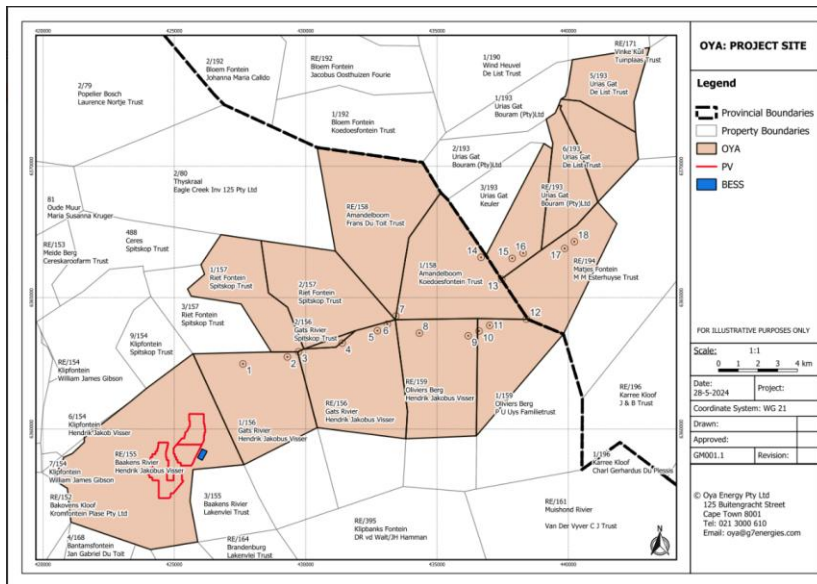


Figure 8: Map of Project boundary

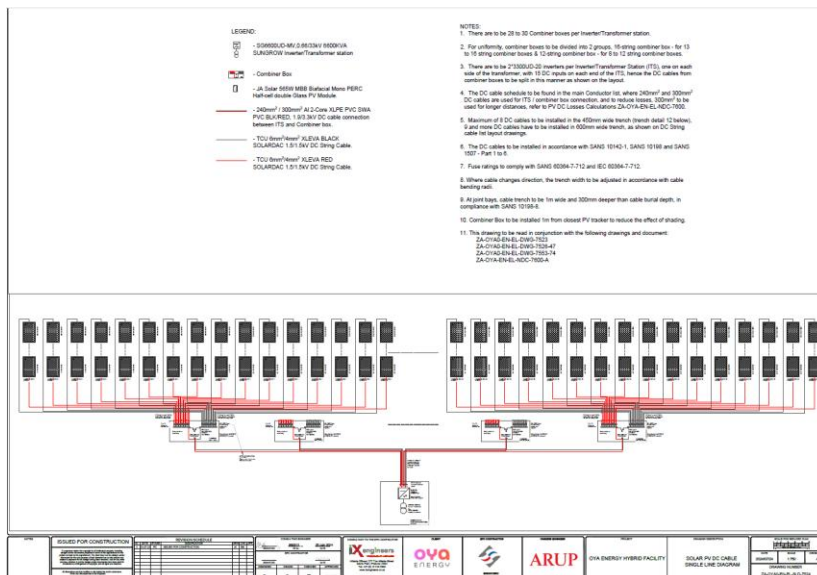


Figure 9: Schematic of Solar PV array

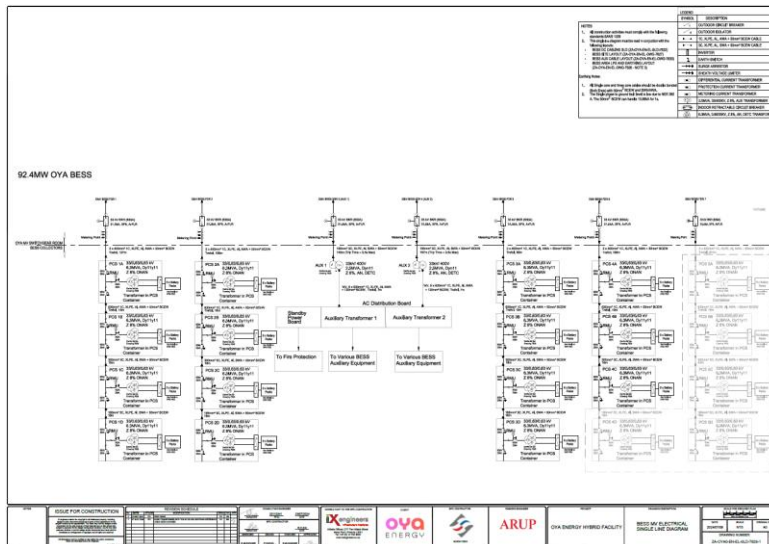


Figure 10: Schematic of the BESS configuration

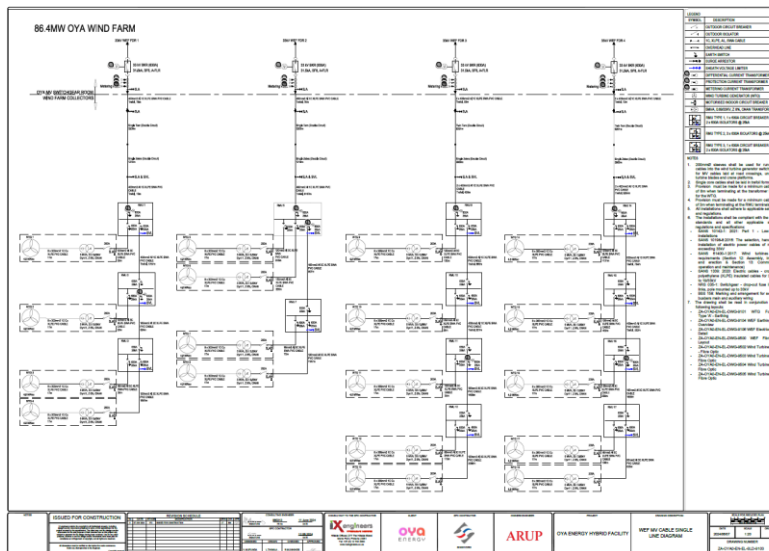


Figure 11: Schematic of the WEF

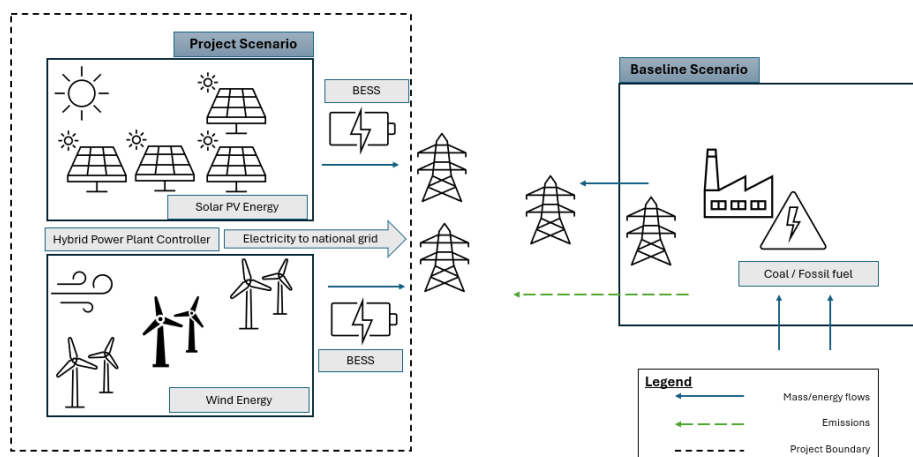


Figure 12: Project boundary, mass & energy flows of Oya Hybrid Project

3.4 Baseline Scenario

The baseline scenario for the Oya Hybrid project is defined as the continuation of current grid operations, where electricity demand is primarily met by existing grid-connected fossil fuel power plants, predominantly coal, and the addition of new fossil fuel-based generation sources. This approach aligns with the requirements of the ACM0002 (Version 22.0) methodology, which states in Section 5.2.1: "If the project activity is the installation of a Greenfield power plant with or without Battery Energy Storage System (BESS), the baseline scenario is electricity delivered to the grid by the project activity that would otherwise have been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in AM TOOL 07 'Tool to calculate the emission factor for an electricity system (Version 07.0)'.

Justification of the Baseline Scenario:

The South African electricity grid is characterized by high carbon intensity due to its heavy reliance on coal-fired power plants, which accounts for 80% of its energy mix. This high reliance on coal positions South Africa among the top greenhouse gas emitters per capita globally.

According to South Africa's Integrated Resource Plan (IRP 2019), coal remains a dominant energy source, projected to account for over 50% of the electricity generation mix through 2030. The IRP outlines significant reliance on existing coal-fired power plants, with limited expansion of renewable energy capacity due to financial, regulatory, and infrastructural barriers. Despite government efforts to increase renewable energy through initiatives like the Renewable Energy

Independent Power Producer Procurement Programme (REIPPPP), the expansion pace is slow, and integration challenges persist, resulting in continued reliance on fossil fuel generation. The South African energy sector faces numerous challenges, including delays in project approvals, difficulties in securing financing, and grid connectivity issues. These barriers significantly impact the deployment of renewable energy projects, validating the assumption that the baseline scenario will continue to be dominated by fossil fuel-based power plants, including both existing and new coal and gas-fired units.

Historical data from Eskom, the primary electricity supplier, indicates a stable grid emission factor of approximately 0.96 tCO₂e/MWh over the past five years, driven by a heavy reliance on coal-fired plants. This trend is expected to continue in the absence of major interventions like the Oya Hybrid project, further supporting the selection of this baseline scenario. The grid emission factor for this baseline is calculated in accordance with the CDM TOOL 07 “Tool to calculate the emission factor for an electricity system” Version 07.0. Option A under the Simple OM approach was selected along with the default weighting for solar projects in the first crediting period. This results in a combined margin factor for the baseline grid of 0.96 tCO₂e/MWh. The steps taken to calculate the grid emission factor as per AM TOOL 07 are outlined in section 4.1 below. §28 – 34 of ACM0002 do not apply to this project, as shown in Table 5 below:

Table 5: Identification of Baseline scenario as per ACM0002

Section number in ACM0002	Applicable to	Applicability to the project
5.2.2	Capacity addition to an existing renewable energy power plant or integration of a BESS to an existing solar photovoltaic or wind power plant/unit	N/A – the project is a Greenfield facility and not a capacity addition to an existing renewable energy power plant. The BESS components are also greenfield and exclusive to the greenfield Oya Hybrid generation facilities.
5.2.3	Baseline scenario for retrofit or rehabilitation or replacement of an existing power plant	N/A – the project is a Greenfield facility and not a retrofit, rehabilitation or replacement of an existing power plant.
5.2.4	Baseline scenario for retrofit of an existing solar photovoltaic or wind	N/A – the project is a Greenfield facility and not a retrofit of an existing facility.

power plant/unit with the integration of BESS

5.2.5	Baseline scenario for pumped storage projects operating in coordination with the renewable energy plants	N/A - the project is a Greenfield facility and is not connected to a Greenfield pumped storage power plant.
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The project activity is the installation of two greenfield power plants with associated BESS component. The baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL 07.

3.5 Additionality

Additionality was demonstrated as stipulated in section 5.3 of ACM0002, using CDM TOOL 01 version 7. In addition, TOOL 27 version 14 and TOOL 24 version 3.1 were applied.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

N/A – the project is located in the Republic of South Africa which is a Non-Annex 1 country and project activities are not mandated by a law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Additionality is demonstrated by proving that the emission reductions achieved by the project would not have occurred in the absence of carbon finance. The following tools were used to evaluate additionality:

- CDM TOOL 01 version 7
- CDM TOOL 27 version 14
- CDM TOOL 24 version 3.1

CDM Guidelines on the assessment of investment analysis (version 5) [10]

The input values used in the investment analysis were valid and applicable at the time of the investment decision taken by the project participant. For this project, financial close was reached in on 13 February 2024 [11], and the investment analysis was therefore carried out with values that were valid in February 2024, as specified in the table below. CDM TOOL 01 version 7 prescribes a stepwise approach for evaluating additionality. These steps were followed as outlined below:

Step 0: Demonstrate whether the proposed activity is the first-of-its-kind.

Although this is one of two projects being developed in parallel that are the first to combine solar PV, wind and battery storage for electricity generation into the South African power grid, this was not considered in the additionality argument, as there are other solar PV and wind projects of a similar scale within the country.

Step 1: Identification of alternatives to the project activity

Step 1 requires that realistic and credible alternatives to the project activity be defined through the following sub-steps:

Sub-Step 1a: Define alternatives to project activity

This step entails defining realistic and credible alternatives available to the project participants or similar project developers that provide outputs or services comparable to the proposed project activities. This is shown in the table below:

Table 6: Alternatives to proposed project activities

Project Alternatives	Applicability to this project
Alternative a: The proposed project activity is undertaken without being registered as a VCS project	Not a realistic scenario: The project proponent could proceed with this project without carbon finance. The project would still proceed as a BESS supplying the South African electricity grid. This is legally and practically an option, but financially it would not be a viable alternative.
Alternative b: Other realistic and credible alternative scenarios to the proposed project activity scenario that delivers electricity with comparable quality, properties and application areas, taking into account examples of scenarios identified in the underlying methodology (ACM0002).	Not a realistic scenario: The supply of stable dispatchable renewable energy to the grid is made possible by the incorporation of a BESS. No other energy storage systems were deemed realistic at the project sites and scale.

Alternative c: Continuation of the current situation (no project activity is undertaken.) This scenario is therefore equal to the baseline scenario where electricity is delivered to the grid through carbon-intensive fossil-fuel derived energy.

Realistic scenario: The grid could continue operating from fossil-fuel powered plants as described in the baseline scenario.

Outcome of sub-step 1a:

Alternative a: The project proponent could not proceed with this project without carbon finance.

Alternative c: Continuation of the current situation (i.e. baseline scenario).

Sub-Step 1b: Consistency with mandatory laws and regulations

The alternative(s) shall be in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution. (This sub-step does not consider national and local policies that do not have legally binding status.) The applicable regulations are mentioned in section 1.15.

The scenarios identified in sub-step 1a include solar, wind, and fossil-fuel derived electricity currently feeding into the South African grid. All three these technology options are in compliance with all legal and regulatory requirements in South Africa. There are no laws prohibiting the implementation of the scenarios identified, and other projects do exist in the country as mentioned in Step 0 above. In addition, all applicable environmental laws have been adhered to as indicated by the approved environmental and water use authorisations.

The applicable laws are listed in Table 4, in section 1.15 of this document.

Outcome of sub-step 1b:

The assessment indicates that scenarios listed above are in compliance with mandatory legislation and regulations in South Africa.

The project proponent can now proceed to step 2 or step 3. Project proponents may also select to complete both steps 2 and 3.

Step 2: Investment analysis

This step entails determining whether the proposed project is not:

- (a) The most economically or financially attractive; or
- (b) Economically or financially feasible, without the revenue from the sale of certified emission reductions (CERs).

The latest version of the “Guidelines on the assessment of investment analysis”, was also considered during this step. Option (b) was selected for analysis which is to conduct a financially attractive option. The following sub-steps were applied:

Sub-step 2a: Determine appropriate analysis method

Determine whether to apply simple cost analysis, investment comparison analysis or benchmark analysis. The scenarios identified in step 1 generate financial benefits and therefore the simple cost analysis is not suited. The alternative of continuing the current scenario does not involve an investment by the project developer and therefore the investment comparison analysis is not suited. The benchmark analysis (option III) was most suitable and therefore used for this analysis.

Sub-step 2b: Apply Investment benchmark analysis

Identify financial indicator:

According to TOOL 01, version 7, either project IRR or equity IRR can be used. The financial indicator identified as most appropriate for this type of project was identified as pre-tax project IRR. This is a parameter that is used commonly within the market. As per the CDM guidelines on the assessment of investment analysis, [10] section III, a pre-tax benchmark is suited to the project IRR calculation. This is due to the impact of loan interest on income tax calculations According to TOOL 27 §16, the applied benchmark shall be appropriate to the type of IRR calculated. Local commercial lending rates or Weighted Average Cost of Capital (WACC) are appropriate benchmarks for a project IRR. Local commercial lending rate was chosen as a suitable benchmark, as obtained from the South African Reserve Bank at the time of investment decision which was 13 February 2024. For the whole of February 2024, the commercial lending rate in South Africa value was 11.75% [12].

Sub-step 2c: Calculation and comparison of financial indicators

The project IRR benchmark analysis was carried out based on the financial assumptions made during the project decision making process. By using a pre-tax benchmark, loan interest and instalments (which have an impact on income tax calculations) are excluded from the calculation. Based on the results, the project IRR is assessed to be less than the selected benchmark. This substantiates that the project would not be able to service debt at South Africa's commercial lending rate and is therefore not financially feasible in the absence of additional income from the sale of carbon credits. (Project IRR for the project activity is less than the Benchmark). Therefore, it can be concluded that the project activity is additional and is not the business-as-usual scenario.

The underlying assumptions are tabulated below:

Table 7: Financial indicators for additionality calculations

Project Parameter	Value	Unit	Source
Total Capacity	128	MW	PPA [13]
Annual generation	544 119	MWh/yr	Production model (average over 20 years)

Total plant Capex ⁵	9 336 515	kZAR	Project Financials
Tariff rate	1 718	ZAR/MWh	PPA (weighted average)
O&M Costs ⁶ including Wind, solar PV and BESS with hybrid power plant controller asset management, planned and preventative maintenance on equipment, corrective maintenance, hybrid performance controller (HPC) and SCADA system, staff and labour costs, Health, Safety and Environmental (HSE) compliance, land leasing costs.	29,900	kZAR/yr	This is the value in the first year as per the project's approved financial model. The cost grows incrementally as per supplier contracts and is inflated annually by projected inflation at the time of investment decision. A portion of the O&M contract is in USD and a portion in ZAR.
ZAR / USD exchange rate	19.02	R/\$	ZAR/USD exchange rate at the time of financial close.
Escalation	4.6	%	South Africa projected inflation at time of investment decision [15].
Insurance	33,500	kZAR/yr	Supplier contracts, inflated annually by South Africa's projected inflation at the time of investment decision.
Salvage value	57,745	kZAR	Decommissioning report
Decommissioning costs net of salvage value	90,347	kZAR	Decommissioning report
Commercial lending rate	11.75	%	SA Reserve Bank [16]

Based on the assumptions above, the pre-tax project IRR would be 9.29% which is below the benchmark of 11.75%

⁵ This includes infrastructure, general and electrical equipment, mechanical equipment, civil works and construction costs, site office establishment, instrumentation and control, environmental approvals, other pre-operative expenses

⁶ A portion of the O&M contract is in ZAR and a portion is in USD. For this calculation, the USD values were converted to ZAR by multiplying with the exchange rate at the time of investment decision. The exchange rate was 19.02 in February 2024 – Reference: [14].

Sub-step 2d: Sensitivity Analysis

A sensitivity analysis was conducted to demonstrate that the project IRR value remains below the benchmark threshold, even when there are variations to the critical input assumptions, as required by the Tool. A range of +10% and -10% was applied. The following variables are considered in the sensitivity analysis:

- Capital cost
- Operations and maintenance costs
- Electricity tariff
- Annual generation

Parameter	Project IRR at -10%	Project IRR at +/- 0%	Project IRR at + 10%
Capex	10.62%	9.29%	8.15%
O&M	9.51%	9.29%	9.07%
Tariff	7.71%	9.29%	10.78%
Annual generation	7.71%	9.29%	10.78%

The results of the sensitivity analysis indicate that none of the input variances put the project at risk of exceeding the benchmark project IRR of 11.75%.

In conclusion, the proposed project activity requires additional revenues from the sale of carbon credits to be economically attractive to project developers and surpass or get closer to the benchmark.

Step 3: Barrier analysis

§28 of TOOL 01 allows for project proponents to proceed to either Step 2 (Investment analysis) or Step 3 (Barrier analysis). Project proponents may also select to complete both Steps 2 and 3. Accordingly, Step 2 is selected. Step 3 Barrier analysis was not deemed necessary and was not considered in this analysis in accordance with the requirements of the Tool.

Step 4: Common practice analysis

The common practice analysis was carried out as stipulated in TOOL 24 version 3.1. This tool outlines a step wise approach, which is described below:

Step 1: Calculate applicable capacity output range as +/- 50% of the total design capacity.

The dispatch capacity for the Oya project is 128 MW and therefore the capacity range for this analysis ranges from 64 MW to 256 MW.

Step 2: Identify similar projects (both VCS and non-VCS) which fulfill all of the following conditions:

Criteria	Applicability to this project	Projects identified
----------	-------------------------------	---------------------

a. The projects are located in the applicable geographical area;	The applicable area is the geographical borders of South Africa.	All the renewable energy projects listed were considered for this analysis.
b. The projects apply the same measure as the proposed project activity;	The measure applied is BESS powered by renewable energy fed into the South African electricity grid.	5,729.70 MW of renewable energy is supplied to the South African grid (extracted from Eskom's website) [17]. This included BESS and non-BESS systems.
c. The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;	The fuel source is a combination of solar and wind with integrated battery storage.	There is only one similar project meeting this criteria in South Africa and it is currently under development [18].
d. The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;	BESS with a hybrid power plant controller powered by Renewable electricity, supplying the grid	BESS with a hybrid power plant controller powered by Solar PV and Wind Energy
e. The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;	64 MW to 256 MW	The capacity of the other project is 75 MW but at the time of submitting this PD it is not yet operational and therefore it falls outside the scope N = 0
f. The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of	There are no comparable, operational projects within South Africa at the time of publishing this PD for global stakeholder consultation.	The other project is not yet commissioned and therefore not valid.

proposed project
activity, whichever is
earlier for the proposed
project activity.

It can therefore be concluded that $N_{ALL} = 0$.

Steps 4 and 5 of this section are not applicable since $N_{ALL} = 0$ (calculating N_{diff}/N_{all} would require dividing by zero).

It can therefore be concluded that the proposed project activity is not common practice within South Africa.

3.6 Methodology Deviations

There are no methodology deviations in this project activity.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The baseline emissions were quantified in accordance with the methodology ACM0002 version 22.0. For this project, it includes electricity delivered to the South African National grid. This was calculated as set out in AM TOOL 07, and the calculation steps are described below.

Baseline Emission Calculation

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Equation 1 (ACM0002 Equation 11)

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Net electricity fed into the grid supplied by the project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using TOOL 07 (tCO₂/MWh)

The results obtained are: 522, 354 tCO₂/ annum, as shown below:

Parameter name	Value	Unit
BE_y	522,354	t CO ₂ / yr
$EG_{PJ,y}$	544,119	MWh/yr
$EF_{grid,CM,y}$	0.96	t CO ₂ /MWh

The project is a greenfield power plant and therefore $EG_{PJ,y}$ is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y}$$

Equation 2 (ACM0002 Equation 12)

Where:

$EG_{PJ,y}$ = Net electricity generation fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$ = Net electricity generation supplied by the project plant to the grid in year y (MWh/yr)

The results obtained are: 544 119 MWh / yr (expected average over 20 years)

Grid Emission Factor Calculation

The grid emission factor for South Africa's national electricity grid was calculated as the weighted average of the system's operating margin (OM) and build margin (BM). The operating margin refers to the group of existing power plants whose current electricity generation would be affected by the proposed renewable energy project activity. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed renewable energy project activity. This calculation is only applicable to grid-connected power plants and does not include off-grid power plants.

In accordance with TOOL 07 The following 6 steps are followed:

- **Step 1:** Identify the relevant electricity systems
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional)
- **Step 3:** Select a method to determine the operating margin (OM)
- **Step 4:** Calculate the operating margin emission factor according to the selected method
- **Step 5:** Calculate the build margin (BM) emission factor
- **Step 6:** Calculate the combined margin (CM) emission factor.

4.1.1 Step 1: Identify the project electricity system

The project electricity system was identified, using Option 2, as: The national grid of South Africa – administered by the national power utility, Eskom Holding SOC Ltd. This is shown schematically in [Figure 13](#) below.

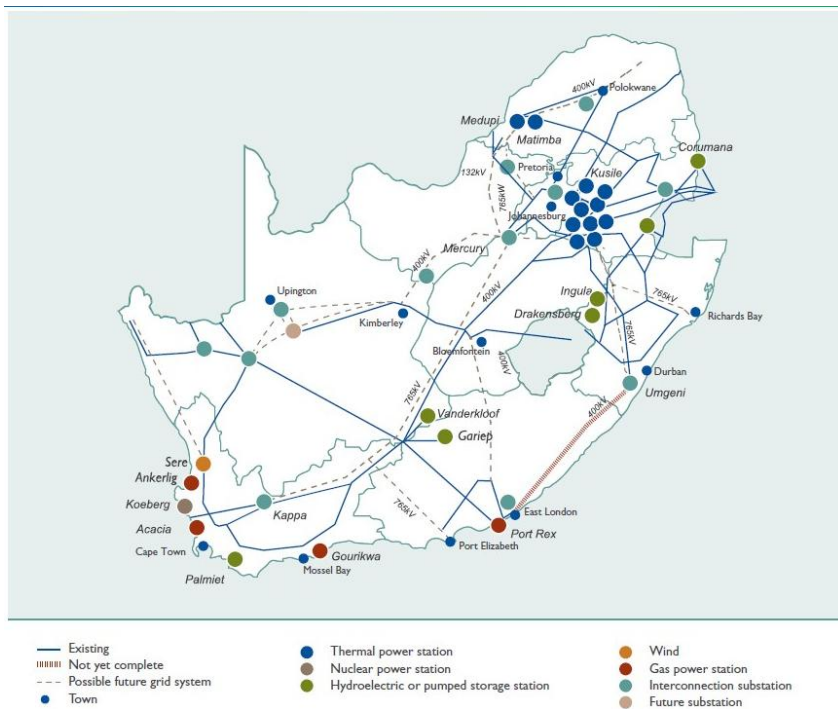


Figure 13: Project electricity system Source: [19]

South Africa's electricity grid is not connected to an electricity system located partially or totally in an Annex I country; therefore, the grid emission factor is not zero.

4.1.2 Step 2: Choose whether to include off-grid power plants in the project electricity system

Only grid connected plants (shown in Table 10). were included in this calculation.

4.1.3 Step 3: Operating Margin method selection

The operating margin (OM) emission factor ($EF_{grid,OM,y}$) was calculated based on the simple OM method (paragraph 38.a in TOOL 07). This was decided based on the following criteria:

- The share of electricity supplied by Low-Cost/Must-Run (LCMR) plants in the last 5 years has ranged from 19% – 23%. It has been consistently lower than 50%, as calculated using TOOL 07 40(a) Approach 1, by [Equation 3](#) below:

$$Share_{LCMR} = average \left[\frac{EG_{LCMR(y-4)}}{total_{y-4}}, \dots, \frac{EG_{LCMR(y)}}{total_y} \right]$$

Equation 3 (TOOL 07, 40 Equation 1)

Where:

- $Share_{LCMR}$ = Share of the low cost/must run resources (%)
- EG_{LCMRy} = Electricity generation supplied by LCMR plants in year y (MWh)
- $total_y$ = Total electricity supplied to the project electricity system in year y (MWh)
- Y = The most recent year for which data is available

The ex-ante option (TOOL 07 paragraph 42(a)) was chosen to determine the simple OM. The 3-year generation-weighted average emission factor was calculated based on the most recent data available at the time of listing the project on Verra's registry. For this project, the most recent data available is 2023. Therefore, the generation weighted average of 2021, 2022, and 2023 was used.

The results obtained are shown in Table 8 below:

Table 8: LCMR Analysis

Parameter name	Value	Unit
$Share_{LCMR: 2023}$	23.1%	Per cent
$Share_{LCMR: 2022}$	21.3%	Per cent
$Share_{LCMR: 2021}$	19.4%	Per cent
$Share_{LCMR: Average}$	21.2%	Per cent
$EG_{LCMRy: 2023}$	49 691	GWh / yr
$EG_{LCMRy: 2022}$	48 092	GWh / yr
$EG_{LCMRy: 2021}$	42 495	GWh / yr
$total_y: 2023$	215 318	GWh / yr
$total_y: 2022$	226 226	GWh / yr
$total_y: 2021$	219 423	GWh / yr

4.1.4 Step 4: Operating margin calculation

The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including LCMR power plants. The following plants were identified as LCMR, and therefore excluded from the OM calculation:

Table 9: LCMR plants serving the project electricity system

LCMR Plant name	Technology	Reason for classification as LCMR
• Gariep • Vanderkloof • Mbashe • First Falls • Ncora • Second Falls	Hydroelectric power stations	Renewable (low generation costs)
• Drakensberg • Ingula • Palmiet	Pumped storage stations	Only used during periods of peak demand (must run)
• Acacia • Ankerlig • Gourikwa • Port Rex • Sere	Gas and liquid turbine peaking stations	Operated only during hours of highest daily, weekly or seasonal loads (must run)
• Koeberg	Wind energy stations	Renewable (low generation costs)
• Biomass • Concentrated solar power • Gas/liquid fuel • Hydroelectric • Landfill • Solar PV • Wind	Nuclear power stations	Nuclear listed as LCMR
	Imports and REIPPP plants	Imports or renewable

The power plants used in the OM calculation are shown in Table 10 below. The simple OM was calculated based on the net electricity generation and the CO₂ emission factor of each power unit (TOOL 07 paragraph 47, option A), as per Equation 4

$$EF_{grid,OMsimple,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 4 (TOOL 07 48, Equation 3)

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (t CO ₂ /MWh)
$EG_{m,y}$	=	Net electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
m power units	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen

The results obtained are: $EF_{grid,OMsimple,y} = 1.004 \text{ tCO}_2/\text{MWh}$

All values used are shown in Table 10.

Table 10: Operating Margin Plant Data

Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
Arnot	10 148 766	9 085 064	7 530 158	1.17	1.16	1.14
Duvha	11 230 400	9 822 489	9 555 266	1.11	1.13	1.13
Hendrina	4 873 506	4 738 179	2 136 620	1.35	1.24	1.20
Kendal	15 106 544	15 139 746	15 750 630	1.02	1.05	1.04
Kriel	11 328 488	12 597 519	10 727 877	1.32	1.28	1.22
Lethabo	21 832 979	22 890 481	21 367 191	1.00	0.97	0.95
Matimba	23 533 126	23 180 336	23 257 630	1.02	1.04	0.99
Majuba	21 857 835	19 823 301	15 540 628	1.19	1.13	1.09
Matla	18 560 297	18 412 664	15 170 449	1.30	1.27	0.99
Tutuka	10 554 111	8 524 715	7 564 337	1.18	1.23	1.22
Camden	2 518 438	6 670 341	7 378 745	1.10	1.16	1.25
Grootvlei	2 992 859	2 445 162	2 416 293	1.21	1.27	1.24
Komati	708 862	608 728	369 330	1.01	1.03	1.06
Kusile	8 553 383	7 992 189	9 119 476	1.01	1.30	1.09
Medupi	19 753 895	22 637 675	23 246 212	0.88	0.92	0.91
Total	183 553 489	184 568 589	171 130 842			

The emission factor of each power unit ($EF_{EL,m,y}$) was determined following option A1 (paragraph 49.a) as shown in Equation 5 below:

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{\sum EG_{m,y}}$$

Equation 5 (TOOL 07 49.(a) Equation 4)

Where:

$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
$FC_{i,m,y}$	=	Amount of fuel type i consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	=	Net calorific value of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO ₂ emission factor of fuel type i in year y (t CO ₂ /GJ)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
i	=	All fuel types combusted in power unit m in year y
y	=	The relevant year as per the data vintage chosen

The emission factor for each power unit ($EF_{EL,m,y}$) is shown in Table 10 above.

The values used in Equation 5 are shown in Table 11 below.

$EG_{m,y}$ was obtained from Eskom's website as well as their annual Integrated Reports [19], [20]

Table 11: Parameters for plant emission factor calculation

Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)	Emission factor coal (tCO ₂ /GJ)	Emission factor fuel oil (tCO ₂ /GJ)
Arnot	5 663 126	5 186 766	4 293 316	45 179.7	37 792.2	36 433.7	21.43	40.40	0.09	0.08
Duvha	6 581 000	5 732 195	5 428 210	71 179.4	39 550.9	34 435.1	20.39	40.40	0.09	0.08
Hendrina	3 024 527	3 024 981	1 486 626	7 722.1	11 012.5	7 013.8	20.51	40.40	0.09	0.08
Kendal	9 272 316	9 362 475	9 395 998	106 402.9	61 174.9	32 683.4	17.87	40.40	0.09	0.08
Kriel	6 694 760	7 838 445	6 865 127	53 949.1	39 516.2	47 877.9	21.64	40.40	0.09	0.08
Lethabo	15 158 029	16 277 939	15 646 686	14 075.3	12 792.0	9 740.7	14.53	40.40	0.09	0.08
Matimba	12 938 911	13 430 751	13 179 578	14 066.5	9 944.3	13 465.6	19.15	40.40	0.09	0.08
Majuba	12 773 092	11 924 693	9 831 151	140 169.2	149 002.7	89 746.1	19.58	40.40	0.09	0.08
Matla	10 694 976	10 842 908	9 536 046	15 923.0	14 520.0	9 179.0	18.19	40.40	0.09	0.08
Tutuka	6 697 281	5 510 214	4 697 013	64 592.8	68 779.3	124 521.3	19.83	40.40	0.09	0.08
Camden	1 569 620	3 932 224	4 129 344	49 016.8	57 014.4	41 321.3	20.45	40.40	0.09	0.08
Grootvlei	1 938 931	1 544 269	1 518 602	21 199.1	61 318.1	19 380.3	20.01	40.40	0.09	0.08
Komati	402 289	335 833	197 954	1 773.0	1 332.0	2 214.0	19.72	40.40	0.09	0.08
Kusile	5 047 912	4 783 403	5 035 930	10 351.0	534 043.2	32 165.0	19.42	40.40	0.09	0.08
Medupi	9 763 379	11 351 788	11 181 866	23 246.2	22 637.7	19 753.9	19.42	40.40	0.09	0.08

4.1.5 Step 5: Calculate the build margin (BM) emission factor

An ex-ante approach (option 1 in TOOL 07 paragraph 72(a)) is selected to calculate the BM emission factor. The most recent information available on units already built for sample group m at the time of submitting the project for validation was used. The procedure followed to determine the sample group of power units (m) used to calculate the build margin is described below. The total electricity generation (AEG_{total}) of all power plants directly supplying the system in 2023 was 184 165.63 GWh.

The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\ units}$) was identified as shown in Table 12. The annual generation of $SET_{5\ units}$ was 35 365.31 GWh in 2023, and this set comprises 19,2% of AEG_{total} .

Table 12: Five most Recent Plants supplying the grid

No.	Plant name	Commissioning Date	Electricity supplied UOM	2023
1	Kusile	2017/08/30	GWh/yr	9,119.48
2	Medupi	2015/08/22	GWh/yr	23,246.21
3	Grootvlei	2008/04/30	GWh/yr	2,416.29
4	Komati	2009/03/31	GWh/yr	369.33
5	Sere Wind	2015/03/01	GWh/yr	214.00
TOTAL				35 365.31

The set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20 per cent of AEG_{total} was determined next, and $SET_{\geq 20\%}$ generated 38 379,02 MWh as shown in Table 13 below, comprised 20,8% of AEG_{total} .

Table 13: Set of power plants that contribute 20% to the Eskom fleet power generation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
1	Kusile	2017/08/30	GWh/yr	9,119.48
2	Medupi	2015/08/22	GWh/yr	23,246.21
3	Ankerlig	2007/03/01	GWh/yr	1,332.49
4	Gourikwa	2007/07/01	GWh/yr	1,681.22
5	Grootvlei	2008/04/30	GWh/yr	2,416.29
6	Komati	2009/03/31	GWh/yr	369.33
7	Sere Wind	2015/03/01	GWh/yr	214.00
TOTAL				38 379.02

SET>20% comprised the larger annual electricity generation and therefore forms the sample set SET_{sample}, but it did contain plants that were commissioned more than 10 years ago and therefore step (d) is followed. units that were commissioned more than 10 years ago were excluded from the set (those shaded grey and CDM, Gold Standard (GS), and VCS projects were included until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system [19]. This resulted in a set that comprised 20% of the project electricity system and therefore steps (e) and (f) were not required. The resulting SET_{sample-CDM} group of power units, m, used to calculate the build margin is shown in Table 14.

Table 14: Plant set for BM Calculation

No.	Plant name	Commissioning Date	UoM	Power Produced in Year: 2023
Eskom Fleet Plants				
1	Kusile	2017/08/30	GWh/yr	9,119.48
2	Medupi	2015/08/22	GWh/yr	23,246.21
3	Sere Wind	2015/03/01	GWh/yr	214.00
CDM, VCS and Gold Standard Plants				
4	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1	2013/03/01	GWh/yr	375.17
5	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/04/01	GWh/yr	94.53
6	Bokpoort CSP (Concentrating Solar Power) Project, South Africa	2013/01/10	GWh/yr	225.86
7	Springbok Grid Connected 55.5 MW Wind Farm (Longyuan Mulilo)	2013/06/01	GWh/yr	130.30
8	Electricawinds 30 MW Wind Project at Riverbank	2013/12/01	GWh/yr	80.30
9	Longyuan Mulilo 1	2012/01/01	GWh/yr	308.14
10	Longyuan Mulilo 2 de Aar, 144 MW	2012/01/01	GWh/yr	497.38
11	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2012/12/01	GWh/yr	878.44
12	Caledon Wind Farm in South Africa	2012/06/01	GWh/yr	406.01
13	Langa Energy Photovoltaic Solar Energy Facility	2012/03/05	GWh/yr	140.45
14	Indwe Wind Project	2012/06/01	GWh/yr	148.09
15	Grid Connected Wind Power Plant in Witberg	2012/02/01	GWh/yr	507.20
16	Red Cap Kouga Wind Farm	2011/11/02	GWh/yr	290.00
	TOTAL CDM, VCS and GS			4081.85
	TOTAL			36,661.5

As the annual electricity generation of this set comprises at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample-CDM} > 0.2 \times AEG_{total}$), this sample group ($SET_{sample-CDM}$) is used to calculate the build margin.

The build margin emission factor was calculated as the generation-weighted average emission factor (tCO₂/MWh) for power units m during 2023, which is the most recent year y for which electricity generation data is available, using [Equation 6](#) ~~Equation 6~~:

$$EF_{grid,BM,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 6 (TOOL 07 77 Equation 15)

Where:

$EF_{grid,BM,y}$	=	Build margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$EG_{m,y}$ (MWh)	=	Net quantity of electricity delivered to the grid by power unit m in year y
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (tCO ₂ /MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

The results obtained are: $EF_{grid,BM,y} = 0.81$

$EG_{m,y}$ and $EF_{EL,m,y}$ were used as shown in Table 10, $EG_{m,y}$ was used as shown in Table 14.

4.1.6 Step 6: Calculate the combined margin (CM) emission factor

The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated as the weighted average CM, as per Equation 7.

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Equation 7 (TOOL 07 85 Equation 16)

As per TOOL 07, the following weighted values (for solar power generation) were used for w_{OM} and w_{BM} :

- $w_{OM} = 0.75$
- $w_{BM} = 0.25$

The results obtained are: $EF_{grid,CM,y} = 0.96$

4.2 Project Emissions

Project emissions are calculated as follows, in accordance with ACM0002 version

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Equation 8 (ACM0002 Equation 1)

Where:

PE_y	=	Project emissions in year y (tCO ₂)
$PE_{FF,y}$	=	Project emissions from fossil fuel consumption in year y (tCO ₂)
$PE_{GP,y}$	=	Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (tCO ₂ e/yr)
$PE_{HP,y}$	=	Project emissions from water reservoirs of hydro power plants (tCO ₂)e/yr)

The project does not involve geothermal power plants or water reservoir of hydro power plants therefore the equation simplifies to:

$$PE_y = PE_{FF,y}$$

This plant will not use fossil fuels for electricity generation, only for emergency backup, therefore $PE_{FF,y}$ can be neglected, in accordance with ACM0002 paragraph 42. Thus project emissions are zero.

4.3 Leakage Emissions

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected, in accordance with ACM0002.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂/yr)

PE_y = Project emissions in year y (t CO₂e/yr)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
01-Oct-2025 to 31-Dec-2025	130, 589	0	0	130, 589	0	130 589
01-Jan-2026 to 31-Dec-2026	522, 354	0	0	522, 354	0	522 354
01-Jan-2027 to 31-Dec-2027	522, 354	0	0	522, 354	0	522 354
01-Jan-2028 to 31-Dec-2028	522, 354	0	0	522, 354	0	522 354
01-Jan-2029 to 31-Dec-2029	522, 354	0	0	522, 354	0	522 354
01-Jan-2030 to 31-Dec-2030	522, 354	0	0	522, 354	0	522 354
01-Jan-2031 to 31-Dec-2031	522, 354	0	0	522, 354	0	522 354
01-Jan-2032 to 30-Sep-2032	391, 766	0	0	391, 766	0	391 766
Total	3,656,480	0	0	3, 656, 480	0	3 656 480

5 MONITORING

5.1 Data and Parameters Available at Validation

The table below provides all data and parameters that are determined or available at validation and remain fixed throughout the project crediting period.

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Simple operating margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 1, as per TOOL 07 “Tool to calculate the emission factor for an electricity system” Data Sources: Records published on Eskom’s Integrated Report [19] and Eskom’s website monthly for each power station to fulfil the Atmospheric Emission License (AEL) requirements [21].
Value applied	1.004
Justification of choice of data or description of measurement methods and procedures applied	The simple OM emission factor was calculated as the generation-weighted average CO ₂ emissions per unit net electricity generation (tCO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units. This means that only Eskom’s coal-based power plants were included in the calculation. The OM was calculated as per option A (paragraph 47.a) of TOOL 07 “Tool to calculate the emission factor for an electricity system”, based on the net electricity generation and a CO ₂ emission factor of each power unit.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ /MWh
Description	Build margin CO ₂ emission factor in year y

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Source of data	Calculated using Equation 6, as per TOOL 07 "Tool to calculate the emission factor for an electricity system" Data source: Records published on Eskom's Integrated Report [19] and Eskom's website monthly for each power station to fulfil the Atmospheric Emission License (AEL) requirements [21] .
Value applied	0.81
Justification of choice of data or description of measurement methods and procedures applied	The BM was calculated as per option 1 (paragraph 72) of TOOL 07 "Tool to calculate the emission factor for an electricity system". An ex-ante approach was followed, based on the most recent information available on units already built. The BM will be updated for the second crediting period based on the most recent information available at that time. Capacity additions from retrofits of power plants are not included in the calculation of the build margin emission factor.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

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Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combine margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 7, as per TOOL 07 "Tool to calculate the emission factor for an electricity system" Data Source:
Value applied	0.96
Justification of choice of data or description of measurement methods and procedures applied	The combined margin (CM) emission factor (EF _{grid,CM,y}) was calculated based on the weighted average CM method (paragraph 81.a) since the data to determine the BM is available. This was done as per equation 16 of TOOL 07 "Tool to calculate the emission factor for an electricity system". Where WOM is equal to: 0.75 and WBM is equal to: 0.25, for the first crediting period and for subsequent crediting periods
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	Share _{LCMR}		
Data unit	Percent (%)		
Description	Share of the low cost/must run resources		
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and records published on Eskom's website [20] for each power station.		
Value applied	Annual LCMR (%) 2021 2022 2023 19.4% 21.3% 23.1%		
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8.		
Purpose of data	To calculate baseline emissions, using Equation 3.		
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.		

Data / Parameter	EG _{LCMR,y}						
Data unit	MWh						
Description	Electricity generation supplied to the project electricity system by the low cost/must run resources in year y						
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and records published on Eskom's website for each power station [22].						
Value applied	<table> <tr> <td>EG_{LCMR,y}: 2023</td><td>49 691 000</td></tr> <tr> <td>EG_{LCMR,y}: 2022</td><td>48 092 000</td></tr> <tr> <td>EG_{LCMR,y}: 2021</td><td>42 495 000</td></tr> </table>	EG _{LCMR,y} : 2023	49 691 000	EG _{LCMR,y} : 2022	48 092 000	EG _{LCMR,y} : 2021	42 495 000
EG _{LCMR,y} : 2023	49 691 000						
EG _{LCMR,y} : 2022	48 092 000						
EG _{LCMR,y} : 2021	42 495 000						
Justification of choice of data or description of	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from						

measurement methods and procedures applied	Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.

Data / Parameter	total _y		
Data unit	MWh		
Description	Total electricity generation supplied to the project electricity system in year y		
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) requirements [21]. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and records published on Eskom's website for each power station		
Value applied	total _y : 2023	215 318	GWh/yr
	total _y : 2022	226 226	GWh/yr
	total _y : 2021	219 423	GWh/yr
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8		
Purpose of data	To calculate baseline emissions, using Equation 3.		
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.		

Data / Parameter	Y
Data unit	Year

Description	The most recent year for which data is available
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) requirements [21].
Value applied	2023
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8.
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.

Data / Parameter	$EF_{grid,OMsimple,y}$
Data unit	tCO ₂ e/MWh
Description	Simple operating margin CO ₂ emission factor in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) requirements [21]. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and records published on Eskom's website for each power station.
Value applied	1.004
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10
Purpose of data	To calculate baseline emissions, using Equation 4.
Comments	The simple OM emission factor is calculated as the generation-weighted average CO ₂ emissions per unit net electricity generation (t CO ₂ /MWh) of all generating power plants serving

the system, not including low-cost/must-run power plants. using Equation 3 under Option A from section 48 in TOOL 07.

Data / Parameter	$EG_{m,y}$			
Data unit	MWh			
Description	Net electricity delivered to the grid by power unit m in year y			
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023, [19] and monthly records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) requirements and from records published on Eskom's website for each power station [21].			
Value applied	Plant Name	Electricity generated (MWh/yr) 2021	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023
	Arnot	10 148 766	9 085 064	7 530 158
	Duvha	11 230 400	9 822 489	9 555 266
	Hendrina	4 873 506	4 738 179	2 136 620
	Kendal	15 106 544	15 139 746	15 750 630
	Kriel	11 328 488	12 597 519	10 727 877
	Lethabo	21 832 979	22 890 481	21 367 191
	Matimba	23 533 126	23 180 336	23 257 630
	Majuba	21 857 835	19 823 301	15 540 628
	Matla	18 560 297	18 412 664	15 170 449
	Tutuka	10 554 111	8 524 715	7 564 337
	Camden	2 518 438	6 670 341	7 378 745
	Grootvlei	2 992 859	2 445 162	2 416 293
	Komati	708 862	608 728	369 330
	Kusile	8 553 383	7 992 189	9 119 476
	Medupi	19 753 895	22 637 675	23 246 212
	Total	183 553 489	184 568 589	171 130 842
Justification of choice of data or description of measurement methods and	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10.			

procedures applied	
Purpose of data	To calculate baseline emissions, using Equation 4.
Comments	The generation-weighted average CO ₂ emissions per unit net electricity generation (tCO ₂ /MWh) of all generating power plants serving the system, not including low-cost/must-run power plants is used to determine the simple OM emission factor, using Equation 3 under Option A from section 48 in TOOL 07.

Data / Parameter	$EF_{EL,m,y}$			
Data unit	tCO ₂ /MWh			
Description	CO ₂ emission factor of power unit m in year y			
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly records published on Eskom's website [for each power station to fulfil the Atmospheric Emission License (AEL) requirements [21]. Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) requirements.			
Value applied	Plant Name	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2021 (tCO ₂ /MWh)
	Arnot	1.17	1.16	1.14
	Duvha	1.11	1.13	1.13
	Hendrina	1.35	1.24	1.20
	Kendal	1.02	1.05	1.04
	Kriel	1.32	1.28	1.22
	Lethabo	1.00	0.97	0.95
	Matimba	1.02	1.04	0.99
	Majuba	1.19	1.13	1.09
	Matla	1.30	1.27	0.99
	Tutuka	1.18	1.23	1.22
	Camden	1.10	1.16	1.25
	Grootvlei	1.21	1.27	1.24

	Komati	1.01	1.03	1.06
	Kusile	1.01	1.30	1.09
	Medupi	0.88	0.92	0.91
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants are shown in Table 10			
Purpose of data	To calculate baseline emissions, using Equation 5.			
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.			

Data / Parameter	$FC_{i,m,y}$			
Data unit	Tonnes			
Description	Amount of fuel type i consumed by power unit m in year y			
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) requirements [21]			
Value applied	Plant Name	Coal usage (t) 2021	Coal usage (t) 2022	Coal usage (t) 2023
	Arnot	5,663,126	5,186,766	4,293,316
	Duvha	6,581,000	5,732,195	5,428,210
	Hendrina	3,024,527	3,024,981	1,486,626
	Kendal	9,272,316	9,362,475	9,395,998
	Kriel	6,694,760	7,838,445	6,865,127
	Lethabo	15,158,029	16,277,939	15,646,686
	Matimba	12,938,911	13,430,751	13,179,578
	Majuba	12,773,092	11,924,693	9,831,151
	Matla	10,694,976	10,842,908	9,536,046
	Tutuka	6,697,281	5,510,214	4,697,013
	Camden	1,569,620	3,932,224	4,129,344
	Grootvlei	1,938,931	1,544,269	1,518,602
	Komati	402,289	335,833	197,954

	Kusile	5,047,912	4,783,403	5,035,930
	Medupi	9,763 379	11,351,788	11,181,866
	Plant Name	Fuel oil usage (t) 2023	Fuel oil usage (t) 2022	Fuel oil usage (t) 2021
	Arnot	45,179.7	37 ,792.2	36,433.7
	Duvha	71,179.4	39,550.9	34,435.1
	Hendrina	7,722.1	11,012.5	7,013.8
	Kendal	106,402.9	61,174.9	32,683.4
	Kriel	53,949.1	39, 516.2	47,877.9
	Lethabo	14,075.3	12,792.0	9,740.7
	Matimba	14,066.5	9, 944.3	13,465.6
	Majuba	140,169.2	149, 002.7	89,746.1
	Matla	15,923.0	14,520.0	9,179.0
	Tutuka	64,592.8	68,779.,3	124,521.3
	Camden	49,016.8	57,014.4	41 321,3
	Grootvlei	21,199.1	61,318.1	19 380,3
	Komati	1,773.0	1 332,0	2,214.0
	Kusile	10,351.0	534,043.2	32,165.0
	Medupi	23,246.2	22,637.7	19,753.9
	As per Table 11			
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants are shown in Table 11			
Purpose of data	To calculate baseline emissions, using Equation 5.			
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.			

Data / Parameter	NCV _{i,y}
Data unit	GJ/tonne
Description	Net calorific value of fuel type i in year y

Source of data	DFFE for coal [24] IPCC default factors for heavy fuel oil , [25]		
Value applied	For coal as per plant in Table 11		
	Plant Name	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)
	Arnot	21.43	40.40
	Duvha	20.39	40.40
	Hendrina	20.51	40.40
	Kendal	17.87	40.40
	Kriel	21.64	40.40
	Lethabo	14.53	40.40
	Matimba	19.15	40.40
	Majuba	19.58	40.40
	Matla	18.19	40.40
	Tutuka	19.83	40.40
	Camden	20.45	40.40
	Grootvlei	20.01	40.40
	Komati	19.72	40.40
	Kusile	19.42	40.40
Medupi	19.42	40.40	
Justification of choice of data or description of measurement methods and procedures applied	Tier 2 NCV (for South Africa) only available for coal. Tier 1 used for other NCVs		
Purpose of data	To calculate baseline emissions, using Equation 5.		
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.		

Data / Parameter	AEG_{total}
Data unit	GWh
Description	Total electricity generation of all power plants directly supplying the system in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL). [21] .

Value applied	184 165.63
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10
Purpose of data	To calculate baseline emissions
Comments	None

Data / Parameter	$SET_{5\text{ units}}$
Data unit	Dimensionless
Description	The set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) [21].
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01) Komati (2009/03/31) Grootvlei (2008/04/40)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 19.2% of AEG_{total} in 2023.

Data / Parameter	$AEG_{set-5-units}$
Data unit	GWh
Description	Annual electricity generation of the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}

Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) [21].
Value applied	35 365.31
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 12
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 19.2% of AEG_{total} in 2023.

Data / Parameter	$SET_{\geq 20\%}$
Data unit	Dimensionless
Description	The set of power units, excluding power units registered as CDM project activities, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) [21].
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Ankerlig (2007/03/01) Gourikwa (2007/07/01) Grootvlei (2008/04/30) Komati (2009/03/31) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generate 20.8% of AEG_{total} in 2023.

Data / Parameter	$AEG_{set \geq 20\%}$
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Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) [21].
Value applied	38 379.02
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 20.8% of AEG_{total} in 2023.

Data / Parameter	SET_{sample}
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5\text{ units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation, and exclude power units that started to supply electricity to the grid more than 10 years ago
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) [21].
Value applied	Kusile (2017/08/30) Medupi (2015/08/22) Sere Wind (2015/03/01)
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13 and Table 14
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 32,579.69 GWh in 2023, comprising 17.7% of AEG_{total} in 2023.

Data / Parameter	$AE G_{set\ sample}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of $AE G_{total}$
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) [21].
Value applied	32,579.69
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 13 and Table 14
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated >20% of $AE G_{total}$ in 2023.

Data / Parameter	$SE T_{sample-CDM}$
Data unit	Dimensionless
Description	The set of power units, (between $SE T_{5\ units}$ and $SE T_{\geq 20\%}$) that comprises the larger annual electricity generation, excludes power units that started producing electricity more than 10 years ago, and including power units registered as CDM project activities, that comprise 20% of $AE G_{total}$
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) [21].
Value applied	Shown in Table 14
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14
Purpose of data	To calculate baseline emissions

Comments	This set of power units generated 36 861.54 GWh in 2023, comprising 20.0% of AEG_{total}
Data / Parameter	$AEG_{SET-sample-CDM}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, excluding power units registered as CDM project activities, and excluding those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2023 [19] and monthly and annual records published on Eskom's website for each power station to fulfil the Atmospheric Emission License (AEL) [21]
Value applied	36 861.54
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 14
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated $\geq 20.0\%$ of AEG_{total} in 2023.
Data / Parameter	w_{OM}
Data unit	Percentage (%)
Description	Weighting of operating margin emissions factor
Source of data	TOOL 07 paragraph 86 (a)
Value applied	75%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM TOOL 07.
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the operating margin emission factor is set at 75% for the first crediting period and for subsequent crediting periods.

Data / Parameter	w_{BM}
Data unit	Percentage (%)
Description	Weighting of build margin emissions factor
Source of data	TOOL 07 paragraph 86 (a)
Value applied	25%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in the CDM TOOL 07.
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the build margin emission factor is set at 25% for the first crediting period and for subsequent crediting periods.

Data / Parameter	BE_y
Data unit	tCO ₂ e/yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 1
Value applied	To be determined ex-post: 522,354
Justification of choice of data or description of measurement methods and procedures applied	Equation provided in ACM0002.
Purpose of data	To calculate baseline emissions, using Equation 1
Comments	None

5.2 Data and Parameters Monitored

The table below provides all data and parameters that will be monitored during the project crediting period. The values provided are used to quantify the estimated reductions and removals for the project crediting period in Section 4 above.

Data / Parameter	$EG_{facility,y}$																		
Data unit	MWh/Year																		
Description	Quantity of net electricity generation supplied by the project plant (i.e. the BESS facility with renewable energy feed from wind and solar PV, hybrid power controller and associated infrastructure) -to the grid in year y																		
Source of data	Electricity reading meters																		
Description of measurement methods and procedures to be applied	Electricity generated is metered automatically, recorded electronically, on a monthly basis, following the Plant's Standard Operating Procedures (SOPs), and checked against the Eskom meter (located at the independent Eskom switching station).																		
Frequency of monitoring/recording	Continuous monitoring and monthly reporting.																		
Value applied	Ex ante values for the first crediting period: <table border="1"> <tr> <td>01-October-2025 to 31-December-2025</td><td>130 589</td></tr> <tr> <td>01-January-2026 to 31-December-2026</td><td>522 354</td></tr> <tr> <td>01-January-2027 to 31-December-2027</td><td>522 354</td></tr> <tr> <td>01-January-2028 to 31-December-2028</td><td>522 354</td></tr> <tr> <td>01-January-2029 to 31-December-2029</td><td>522 354</td></tr> <tr> <td>01-January-2030 to 31-December-2030</td><td>522 354</td></tr> <tr> <td>01-January-2031 to 31-December-2031</td><td>522 354</td></tr> <tr> <td>01-January-2032 to 30-September-2032</td><td>391 766</td></tr> <tr> <td>Total</td><td>3,656,480</td></tr> </table>	01-October-2025 to 31-December-2025	130 589	01-January-2026 to 31-December-2026	522 354	01-January-2027 to 31-December-2027	522 354	01-January-2028 to 31-December-2028	522 354	01-January-2029 to 31-December-2029	522 354	01-January-2030 to 31-December-2030	522 354	01-January-2031 to 31-December-2031	522 354	01-January-2032 to 30-September-2032	391 766	Total	3,656,480
01-October-2025 to 31-December-2025	130 589																		
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01-January-2030 to 31-December-2030	522 354																		
01-January-2031 to 31-December-2031	522 354																		
01-January-2032 to 30-September-2032	391 766																		
Total	3,656,480																		
Monitoring equipment	Electricity meter, (to be confirmed during installation and commissioning), meters will be located at the Switching Station which serves as the grid tie-in point. There will be a meter (with a backup as a check and failsafe. The national utility will also have a meter located in the switching station at their tie in point as a check meter. All these meters are located after the hybrid power controller on either side of the grid tie-in point,																		
QA/QC procedures to be applied	Internal Operational QA/QC for meter readings have been developed and will meet globally acceptable industry norms. Specifics of the QA/QC will be codified during the																		

	implementation and commission of the plant and included at verification.
Purpose of data	Calculation of project emissions
Calculation method	Net electricity supplied to the grid by the project plant (i.e. the BESS facility and associated infrastructure) is directly measured and reported, not calculated.
Comments	No additional comments

Data / Parameter	PE_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 8
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	0
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Calculation of project emissions
Calculation method	Equation 8
Comments	For all renewable energy power generation project activities, emissions due to fossil fuels for the backup generator can be neglected. This project does not involve geothermal power plants or water reservoir of hydro power plants. Therefore, project emissions for this renewable energy power generation project will be zero.

Data / Parameter	ER_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 9
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	To be calculated ex-post – Ex-ante value is 522 354 per annum
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.
Purpose of data	Calculation of project emissions
Calculation method	Equation 9
Comments	The emission reductions determined at validation apply estimates of likely emission reductions from the proposed project activity during the crediting period, using available ex ante values. These will be revised during each verification using the actual data recorded (where applicable).

5.3 Monitoring Plan

Understanding that the project is still in design and implementation phase and has not been constructed and final equipment and instrumentation is still to be installed and implemented. This is the best understanding at the time of validation and will be amended and corrected in an “as-built” version of the plant.

Methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters.

Electricity generated by the facility will be monitored internally using an integrated Distributed Control System (DCS), as part of an overall Supervisory Control and Data Acquisition (SCADA) system which will be used for operational control and reporting. Apart from reporting for this project, the data will also be used to bill the national utility for electricity supplied to the grid. All data and reports will be stored and backed up in accordance with the DCS system design.

Measurement data will be managed by the project's DCS. The measuring equipment will be installed in three steps:

- Calibration under standard temperature and pressure conditions
- Installation and electrical connection by a certified external contractor
- Commissioning and final test under project conditions by the internal instrumentation team

All data will be stored on a server on-site and also loaded onto a firewall protected cloud storage system. All measuring equipment will strictly comply with their specified maintenance plans and calibration certificates will be kept on record.

As per the applied methodology, electricity generation will be measured onsite at the outgoing substation, adjacent to the power generating units. The supply of outgoing electricity will be managed by the hybrid power plant controller which is linked to the SCADA system. This will be monitored on a continuous basis, with a monthly report being aggregated and reported upon to management and in accordance with the PPA. A check meter, owned by the national utility (Eskom) located at the Eskom Switching Station which serves as the grid tie-in point. Meters at this location will have a back-up meter installed as a failsafe.

The meters on site will be calibrated in accordance with the manufacturer's specifications and will form part of the planned maintenance schedule for both checking and calibration. Instrument accuracy checks and verification will be conducted by Competent Instrumentation technicians and records will be kept on site. Calibration of the energy meters will be conducted by competent technicians under SANAS accreditation. The applicable national standard to be applied is SANS 474 and will form part of planned maintenance schedules. This defines the required frequency of calibration in §4.7.4 for meters greater than 10MVA as being every 5 years and less than 10MVA as being every 10 years.

Organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.

Data collection, consolidation and results analysis will be undertaken by a dedicated data and instrumentation team who are adequately trained and competent. They will be responsible for the maintenance (planned and unplanned), checking and arranging calibration for plant equipment. They will also be responsible for maintaining records of calibration and procurement of any instrumentation equipment that may be required for the operation of the plant. All instrument technicians will be suitably competent and records of their training and qualifications will be maintained by the Human Resources department. Only SANAS qualified organisations and personnel will conduct calibrations.

Procedures for internal auditing and QA/QC

Metered electricity production will be recorded by the plant on the DCS system. This will be checked against the Eskom meter at the independent switching stations. Any discrepancies will immediately trigger fault finding and corrective action procedures. There is also a backup meter both on site and at the Eskom stations which are used as an additional failsafe for QA/QC purposes. All reported electricity generation values will be checked by both project personnel and Eskom personnel prior to billing and payment.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

No commercially sensitive information is presented in this document

[6]APPENDIX 2: BIBLIOGRAPHY

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